Gene Expression Pattern Matching

Evaluating Similarity Metrics for NMF Constraint Patterns

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Motivation

- **Goal:** Identify genes whose expression over time matches biologically observed constraint patterns
- **Data:** $\sim$17,856 genes across 4 timepoints (0, 3, 6, 9) in 4 clusters, raw and log-transformed
- **Problem with NMF:** All factors are non-negative, so NMF ranks genes by coefficient magnitude, not shape match. Up-regulation and down-regulation are conflated.
- **Our approach:** Compare 6 similarity metrics that directly measure gene–pattern shape similarity

Constraint Patterns

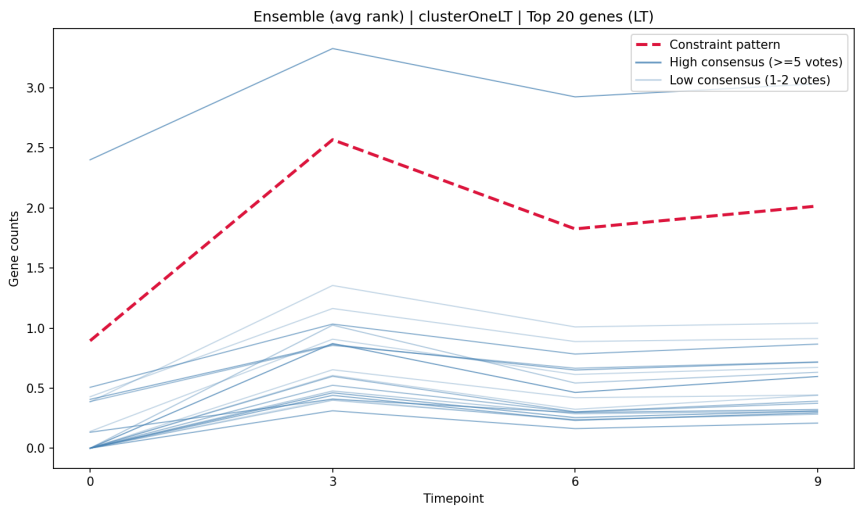
Cluster	**t=0**	**t=3**	**t=6**	**t=9**	**Shape**
One	1.45	12.0	5.21	6.51	Spike at t=3
Two	0.371	0.803	1.22	1.78	Steady increase
Three	0.041	0.0	0.037	0.035	Near-flat, dip at t=3
Four	0.009	0.638	0.018	0.024	70× spike at t=3

All data min-max normalized to $(0, 1]$ before comparison. Both raw and log-transformed (LT) versions analyzed.

The 6 Metrics at a Glance

Metric	Measures	Ranking	Key property
Pearson	Shape (mean-centered)	Higher = better	Scale & shift invariant
Cosine	Vector angle (raw)	Higher = better	No mean-centering
DTW	Time-aligned distance	Lower = better	Allows temporal warping
Fréchet	Max curve deviation	Lower = better	Worst-point sensitive
MSE	Avg squared error	Lower = better	Simplest distance
sMAPE	Symmetric % error	Lower = better	Relative scale

Results: clusterOne LT — Modest Consensus



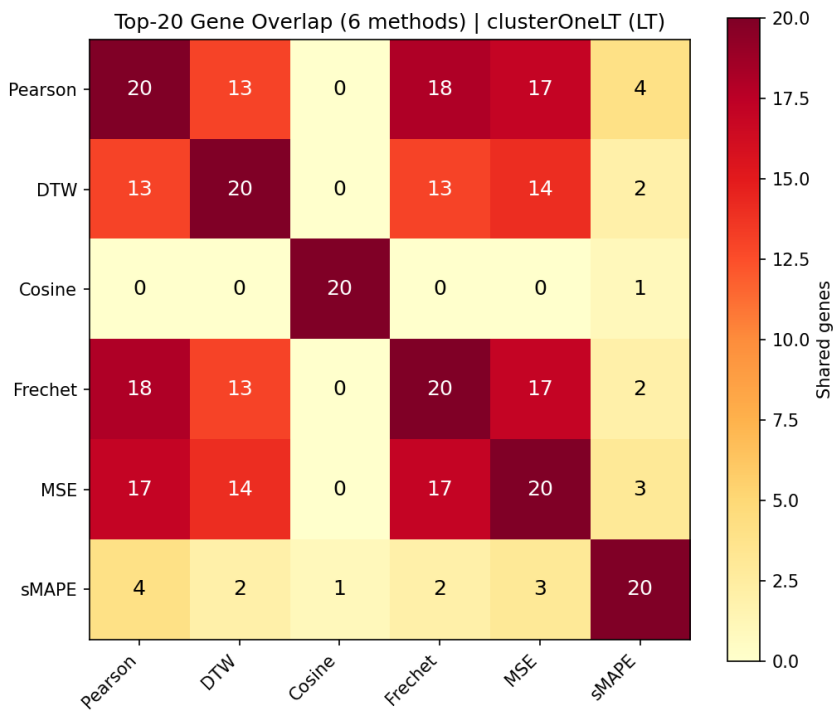
2 genes achieve 5/6 votes:

NACA, AC008875.3

4 additional genes at 4/6:
CCDC57, RTRAF, MDFIC, CTDSPL2

Spike-at-$t = 3$ pattern — distance/correlation methods converge, but no unanimous calls.

Results: clusterOne LT — Pearson/Fréchet/MSE Core



Pearson–Fréchet–MSE: 17–18/20 overlap
DTW joins at 13–14/20
Cosine isolated: 0/20
sMAPE isolated: 1–4/20

Cosine fails even on log-transformed data — magnitude bias persists.

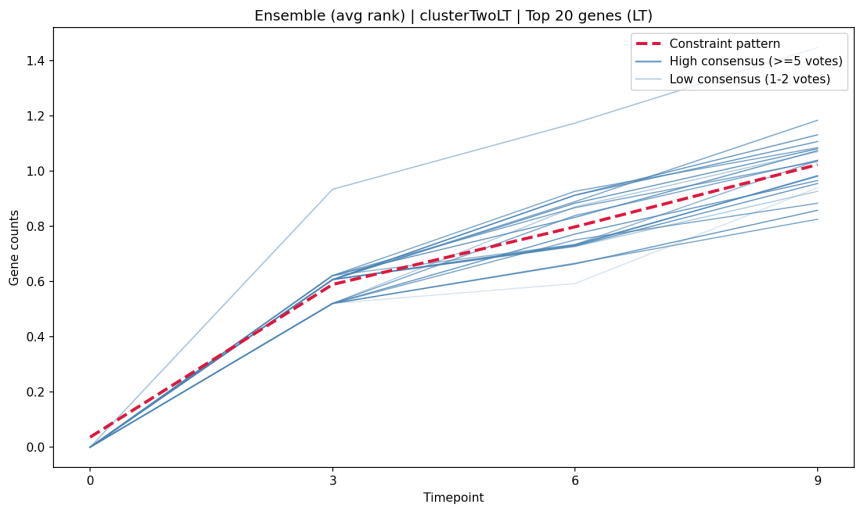
sMAPE penalizes relative error on near-zero values.

Ensemble Rankings — clusterOne (LT)

Rank	Gene	Votes	Methods
1	NACA	**5/6**	Pearson, DTW, Fréchet, MSE, sMAPE
2	AC008875.3	**5/6**	Pearson, DTW, Fréchet, MSE, sMAPE
3	CCDC57	4/6	Pearson, DTW, Fréchet, MSE
4	RTRAF	4/6	Pearson, DTW, Fréchet, MSE
5	MDFIC	4/6	Pearson, DTW, Fréchet, MSE

NACA and AC008875.3 are the highest-confidence candidates for the clusterOne LT spike pattern — consistent across 5 of 6 methods.

Results: clusterTwo LT — Best Agreement



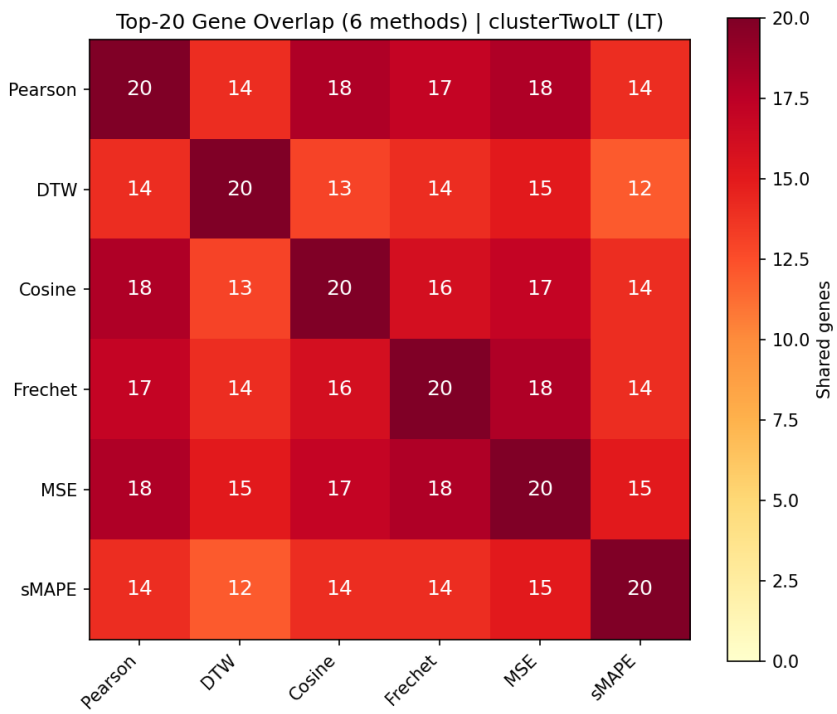
9 of top 11 genes achieve unanimous 6/6 votes:

RAB14, SYNGR1, ARHGAP25,
HNRNPH3, BSG, CAPN2,
NOL7, ZC3H15, PRDX6

Steady-increase pattern is easiest to match — all methods converge.

Highest-confidence candidates
for biological follow-up.

Results: clusterTwo LT — All Methods Converge



Most balanced cluster overall.

Pearson–MSE: 18/20
Pearson–Cosine: 18/20
Fréchet–MSE: 18/20

Even sMAPE agrees: 12–15/20
across all methods — no zero-value failure here.

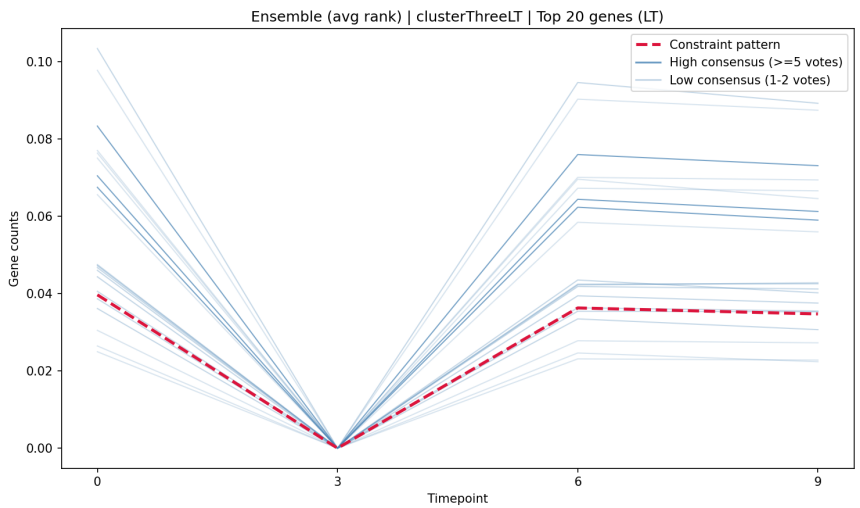
Steady-increase pattern is robust
across all 6 metrics.

Ensemble Rankings — clusterTwo (LT)

Rank	Gene	Votes	Methods
1	RAB14	**6/6**	All methods
2	SYNGR1	**6/6**	All methods
3	ARHGAP25	**6/6**	All methods
4	HNRNPH3	**6/6**	All methods
5	BSG	**6/6**	All methods

9 of the top 11 genes are unanimous 6/6 across all methods — highest-confidence candidates for TCR pathway involvement.

Results: clusterThree LT — 5 of 6 Methods Agree



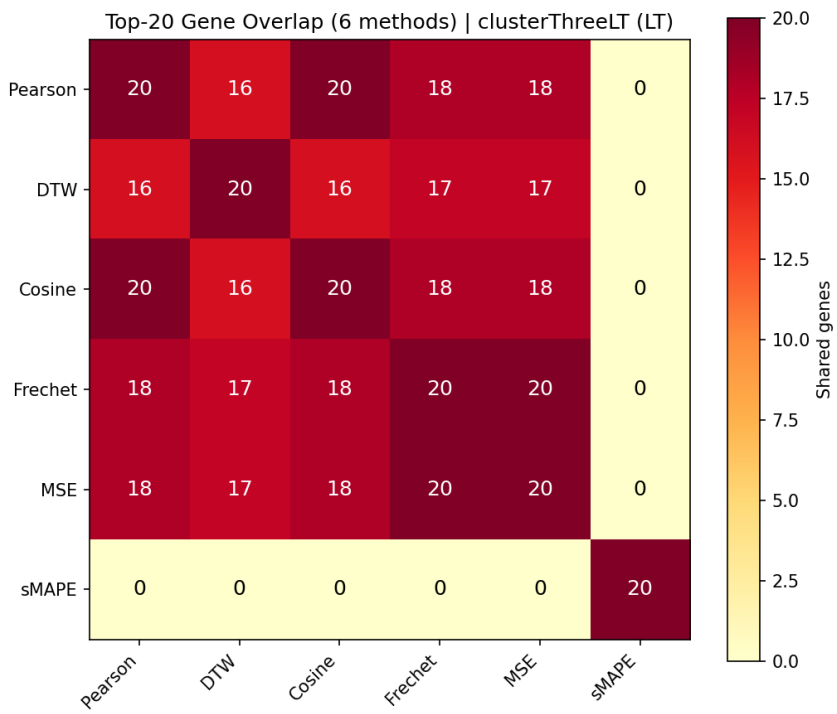
3 genes achieve 5/6 votes:

TEDC1, CTBP1-AS, PGAP1

Pearson, DTW, Cosine, Fréchet, MSE all converge on the near-flat dip pattern.

sMAPE blocked by zero at $t = 3$: $\frac{2|0-g_i|}{|0|+|g_i|} = 2$ regardless of g_i.

Results: clusterThree LT — Strong Consensus (Except sMAPE)



Pearson–Cosine: **20/20** (perfect)
Fréchet–MSE: **20/20** (perfect)
Cross-pairs: 16–18/20

sMAPE: **0/20** with everything (zero-value failure at $t = 3$).

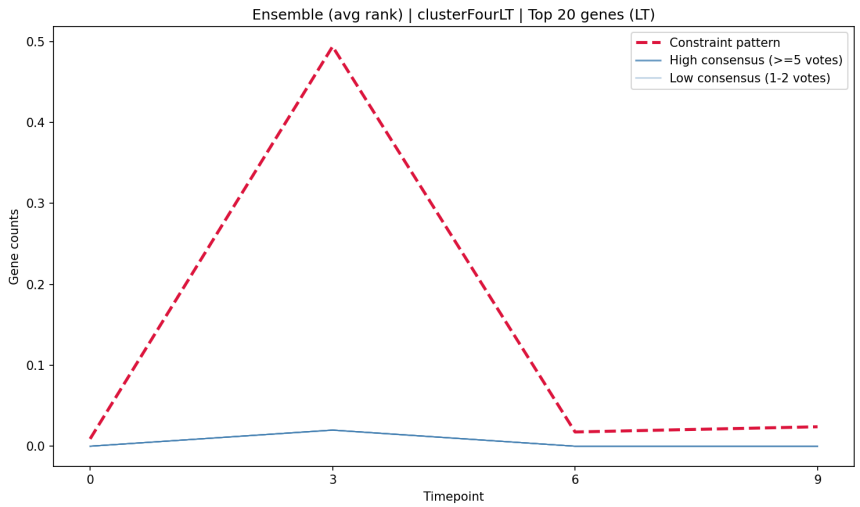
Validates that 5 of 6 methods capture the same biology.

Ensemble Rankings — clusterThree (LT)

Rank	Gene	Votes	Methods
1	TEDC1	**5/6**	Pearson, DTW, Cosine, Fréchet, MSE
2	CTBP1-AS	**5/6**	Pearson, DTW, Cosine, Fréchet, MSE
3	PGAP1	**5/6**	Pearson, DTW, Cosine, Fréchet, MSE

TEDC1, CTBP1-AS, and PGAP1 are the high-confidence consensus genes for clusterThree LT (sMAPE excluded due to the zero-at-t=3 failure).

Results: clusterFour LT — The Hard Case



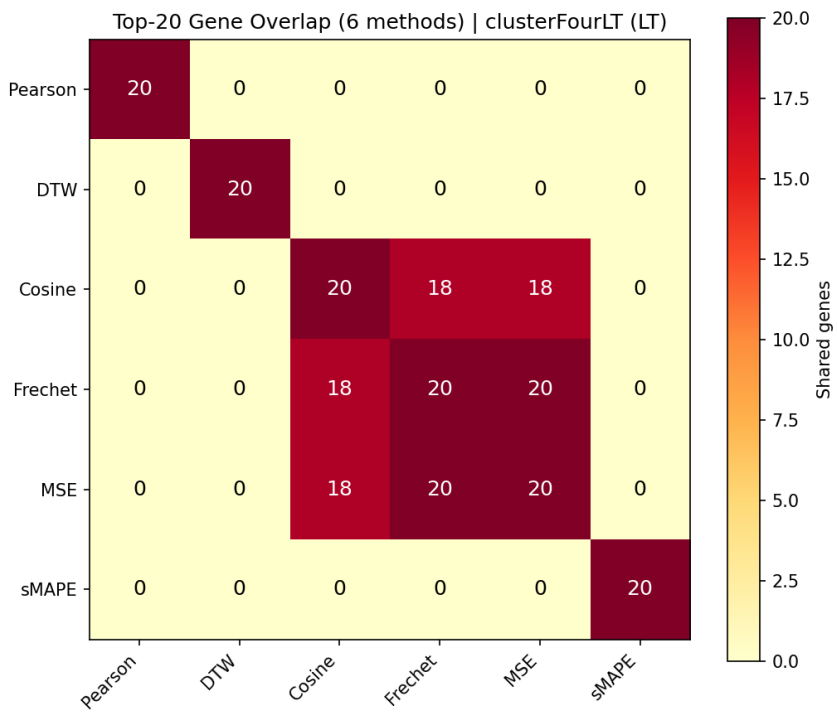
$70\times$ spike at $t = 3 \Rightarrow$ effectively a delta function.

Zero genes achieve $\geq 2/6$ votes in the ensemble top-20.

Methods split into 3 disjoint groups:

- Cosine/Fréchet/MSE (magnitude)
- Pearson (relative shape)
- DTW, sMAPE (each isolated)

Results: clusterFour LT — No Shape Consensus



Cosine–Fréchet–MSE: 18–20/20
(all dominated by $t = 3$ magnitude)

Pearson: 0/20 with everything
(scale-invariant, finds *relative* peaks)

DTW: 0/20 with everything
sMAPE: 0/20 with everything

Needs denser time sampling
to resolve shape vs magnitude.

Ensemble Rankings — clusterFour (LT)

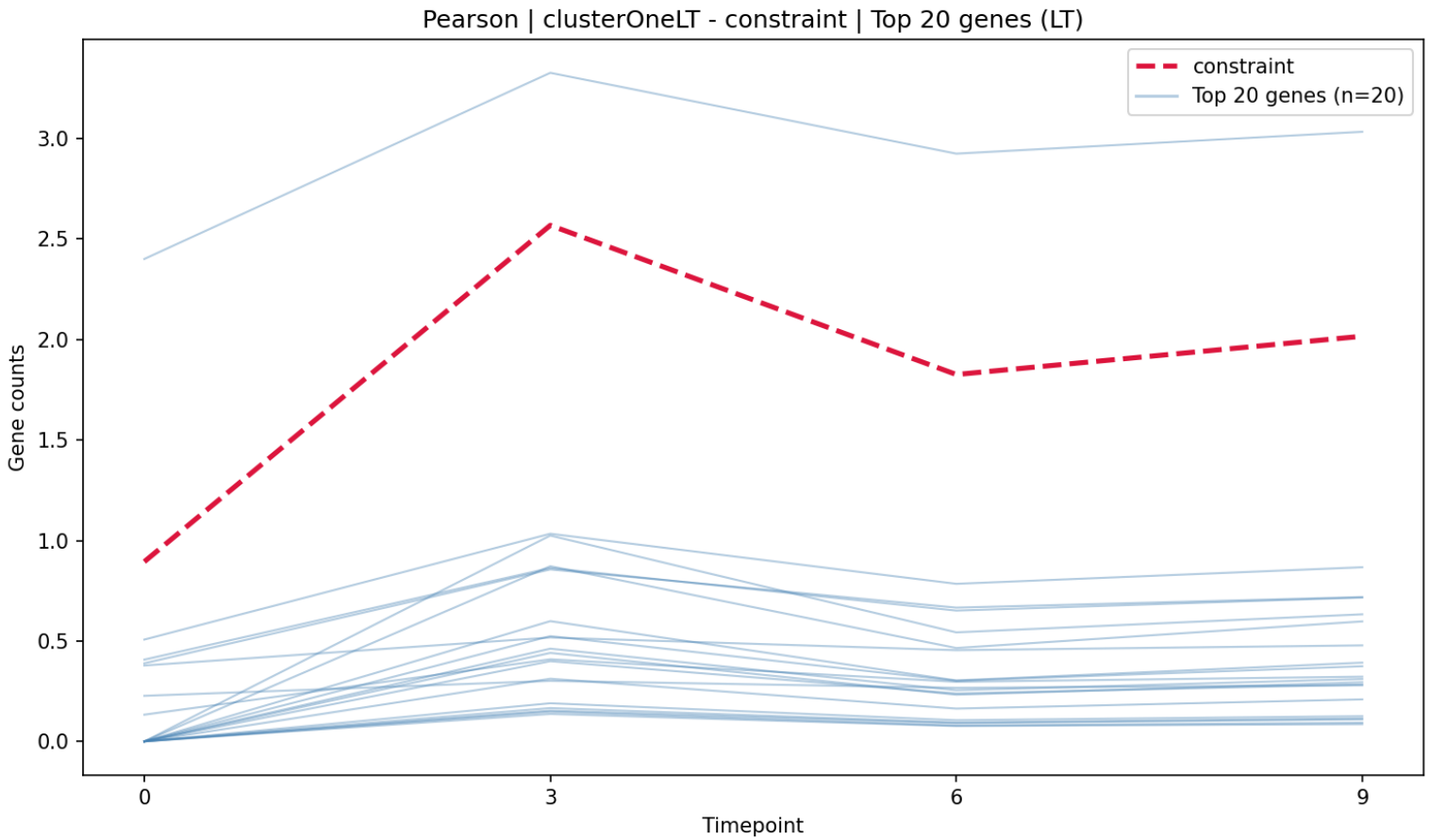
Rank	Gene	Votes	Methods
1	AC068888.2	0/6	—
2	AC018638.6	0/6	—
3	SLC16A5	0/6	—
4	TPH1	0/6	—
5	CNN3	0/6	—

No consensus. Top genes by average rank do not appear in *any* single method's top-20 — the 70× spike defeats the voting ensemble. clusterFour requires denser temporal sampling.

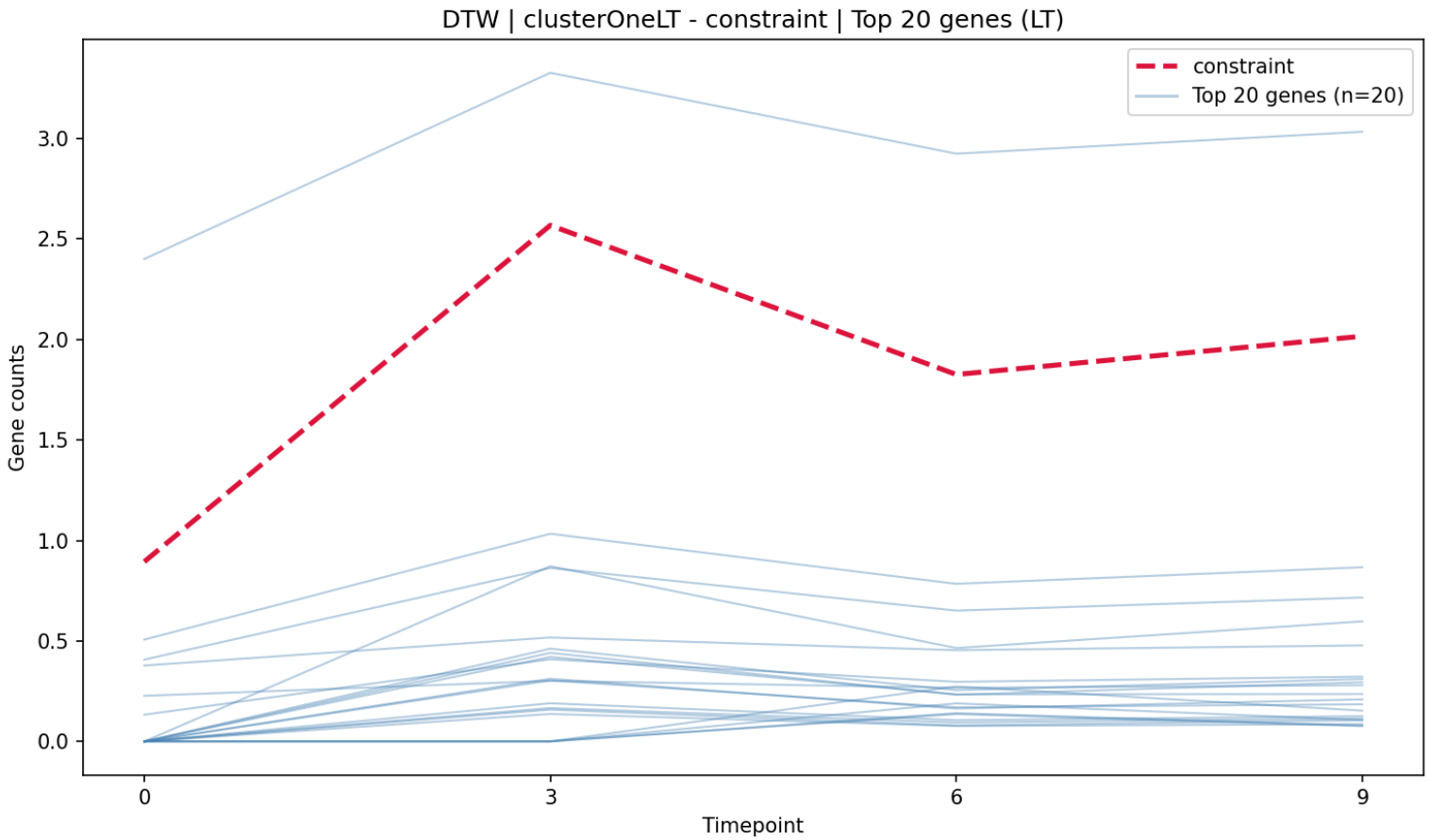
Appendix

All Methods — Log-Transformed Clusters

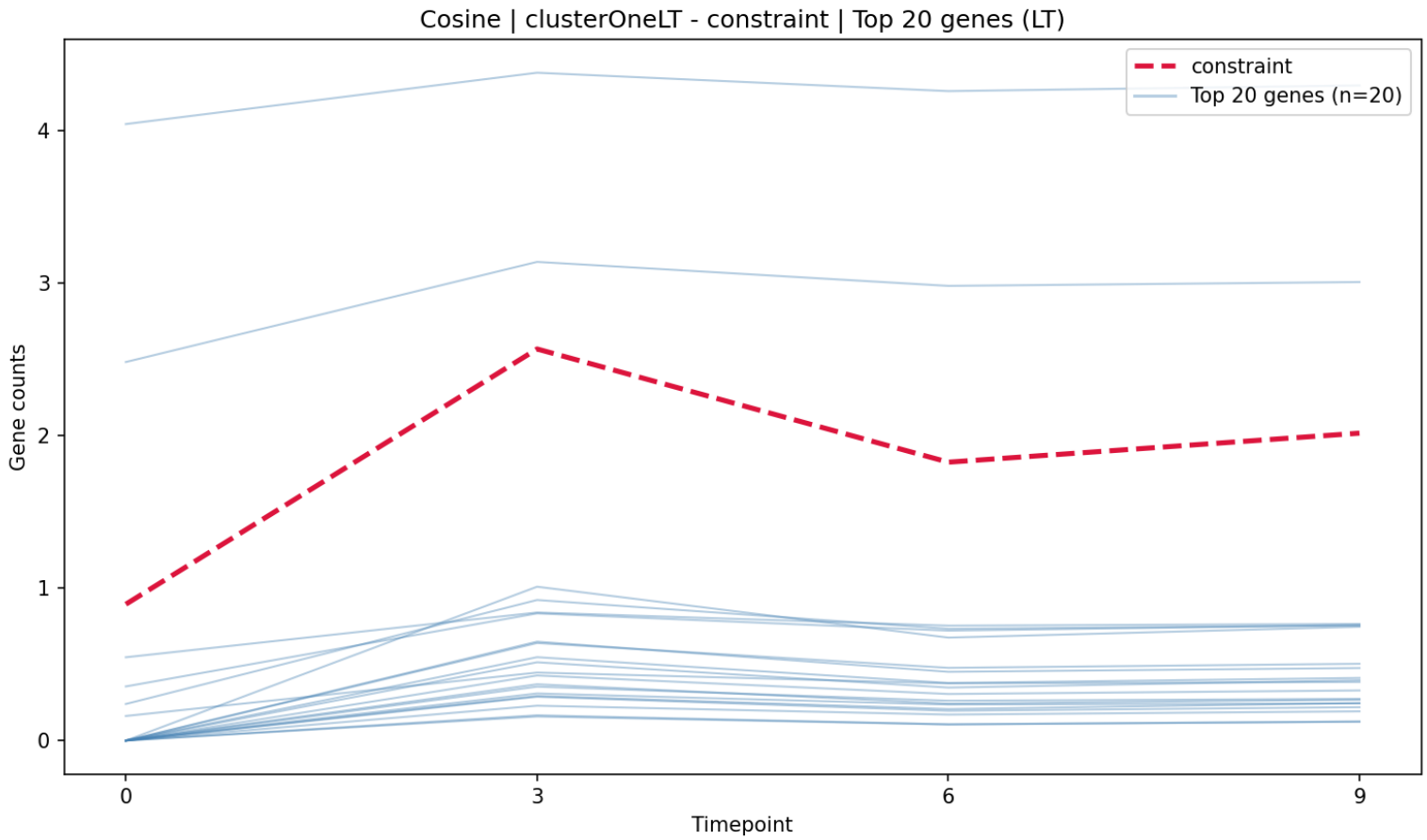
clusterOne LT — Pearson



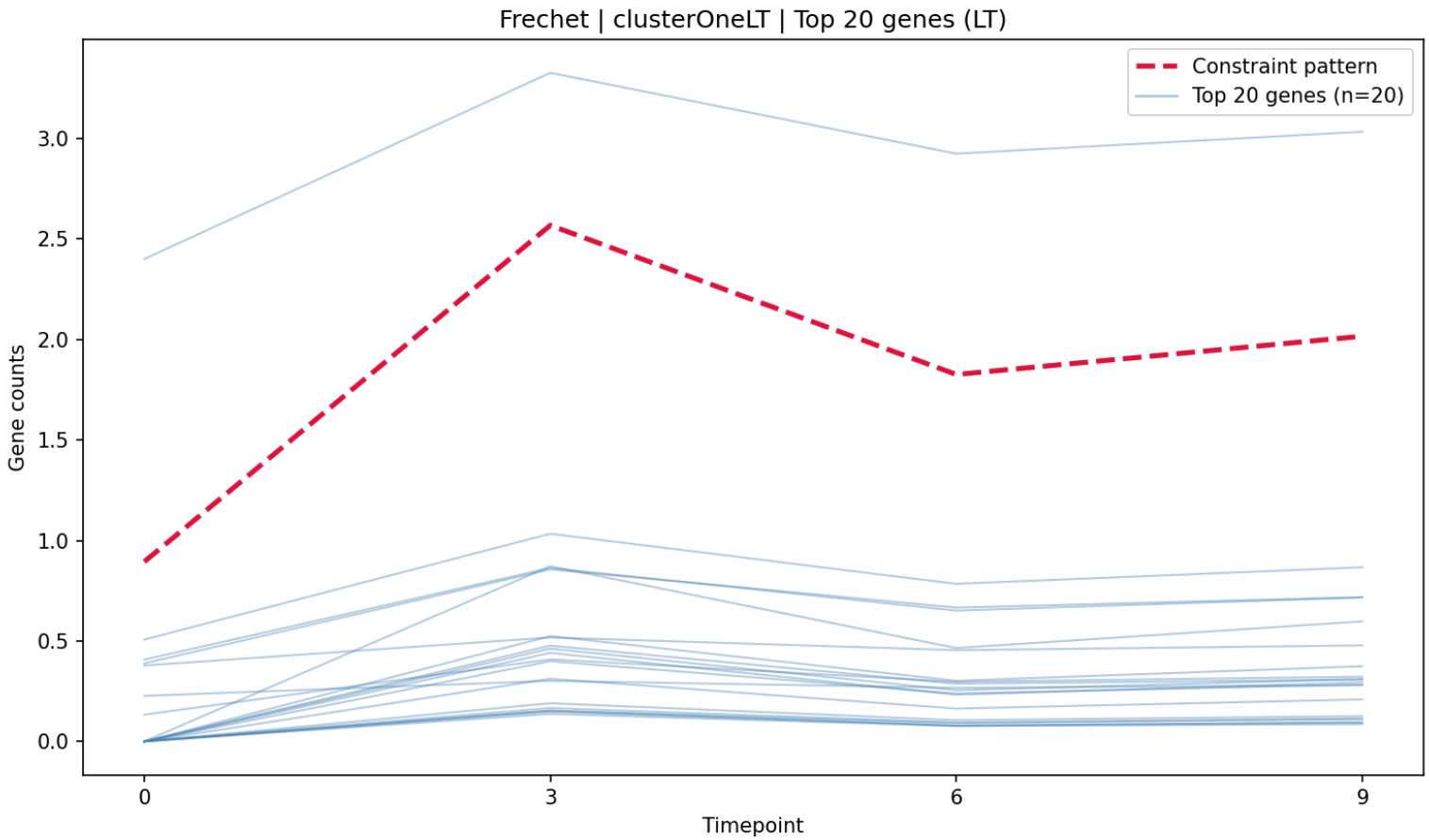
clusterOne LT — DTW



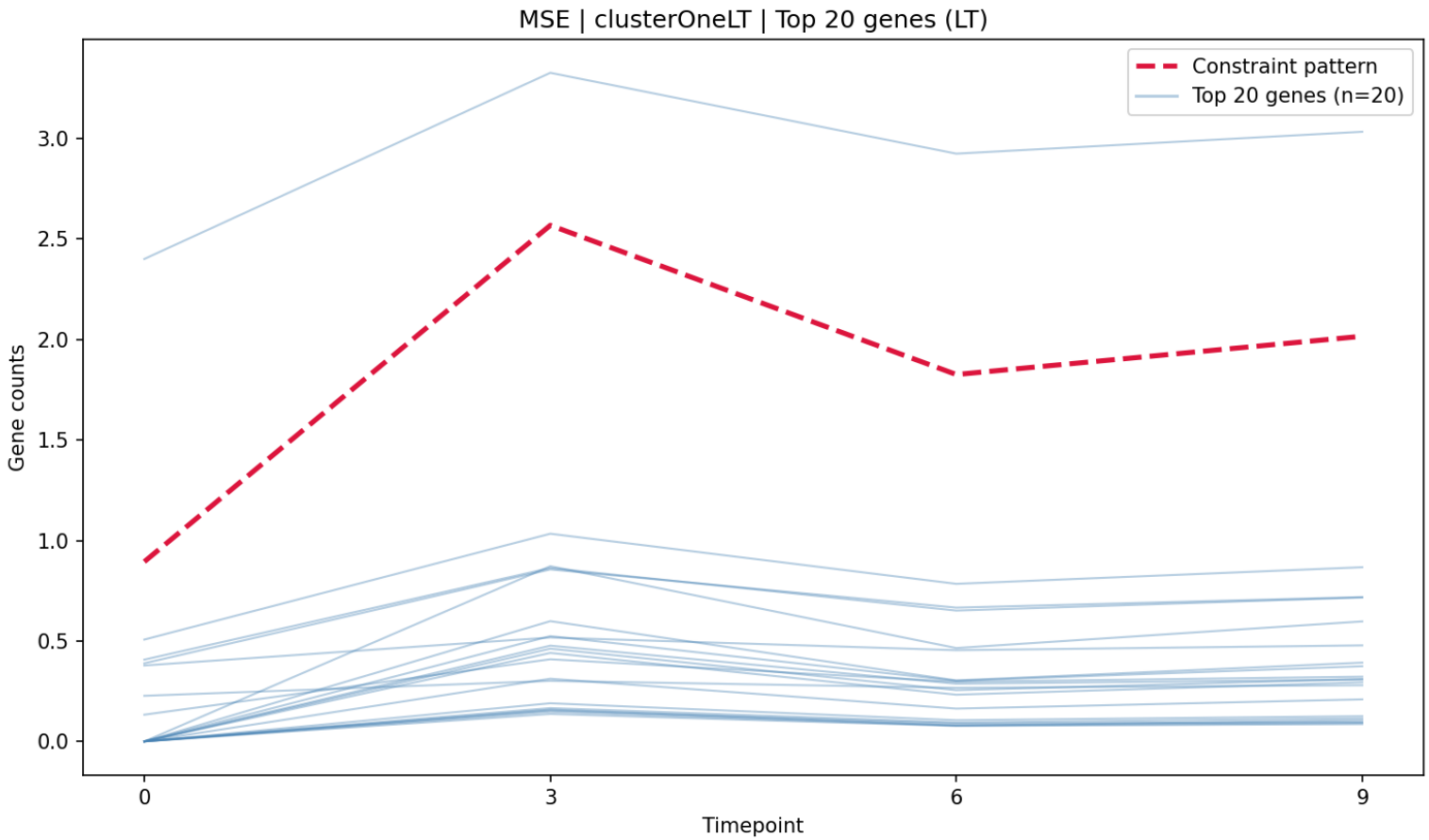
clusterOne LT — Cosine



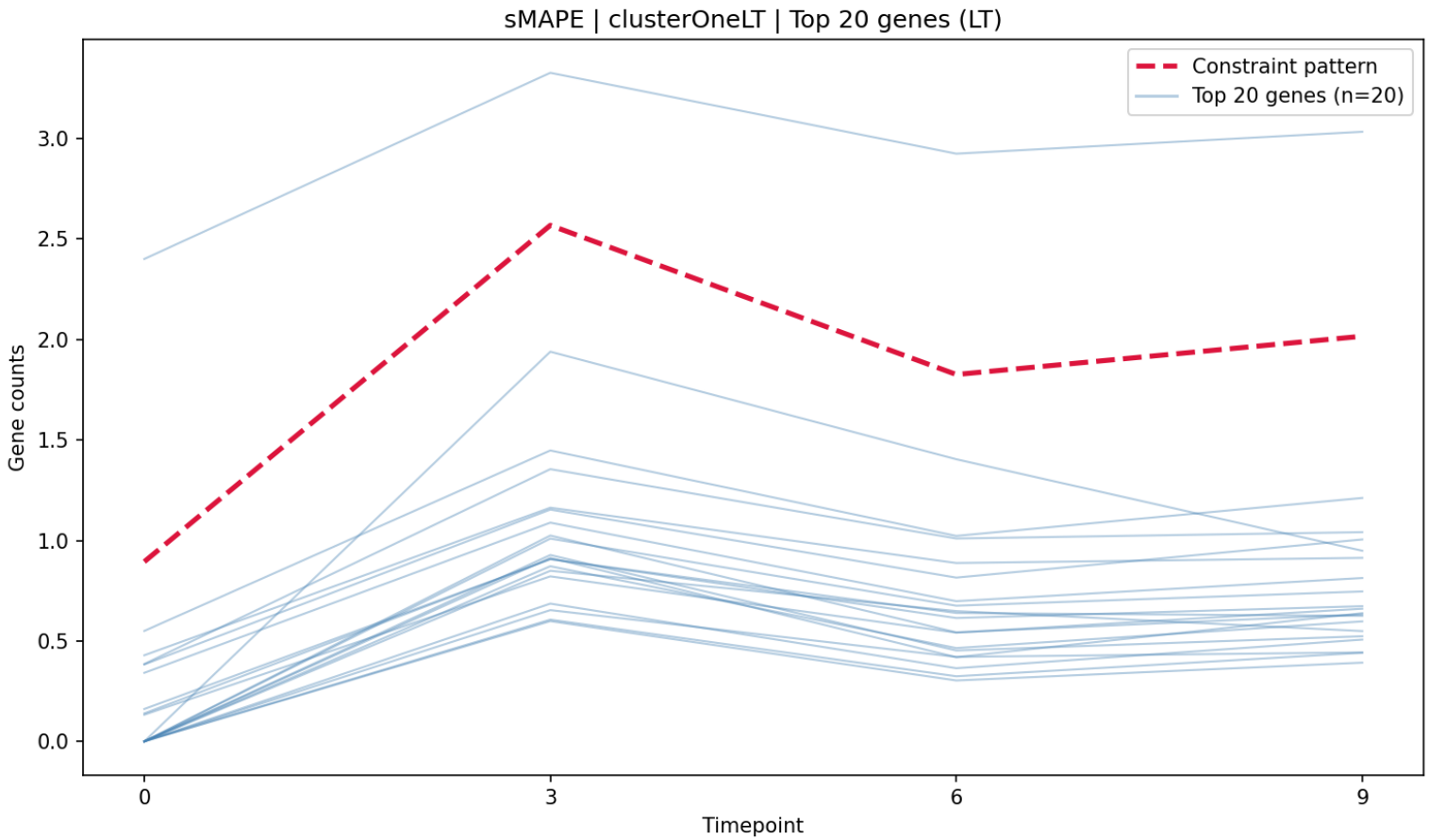
clusterOne LT — Fréchet



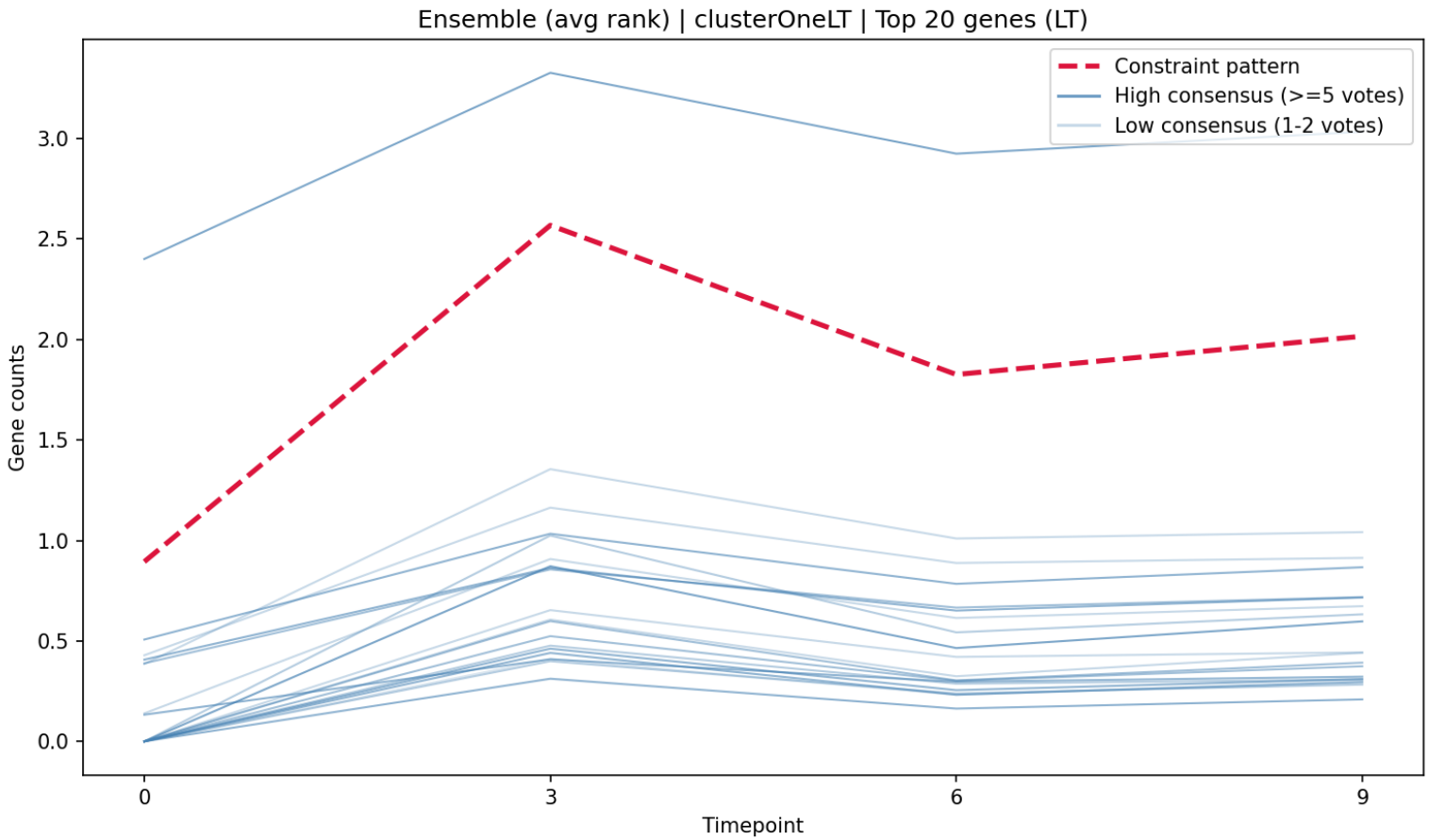
clusterOne LT — MSE



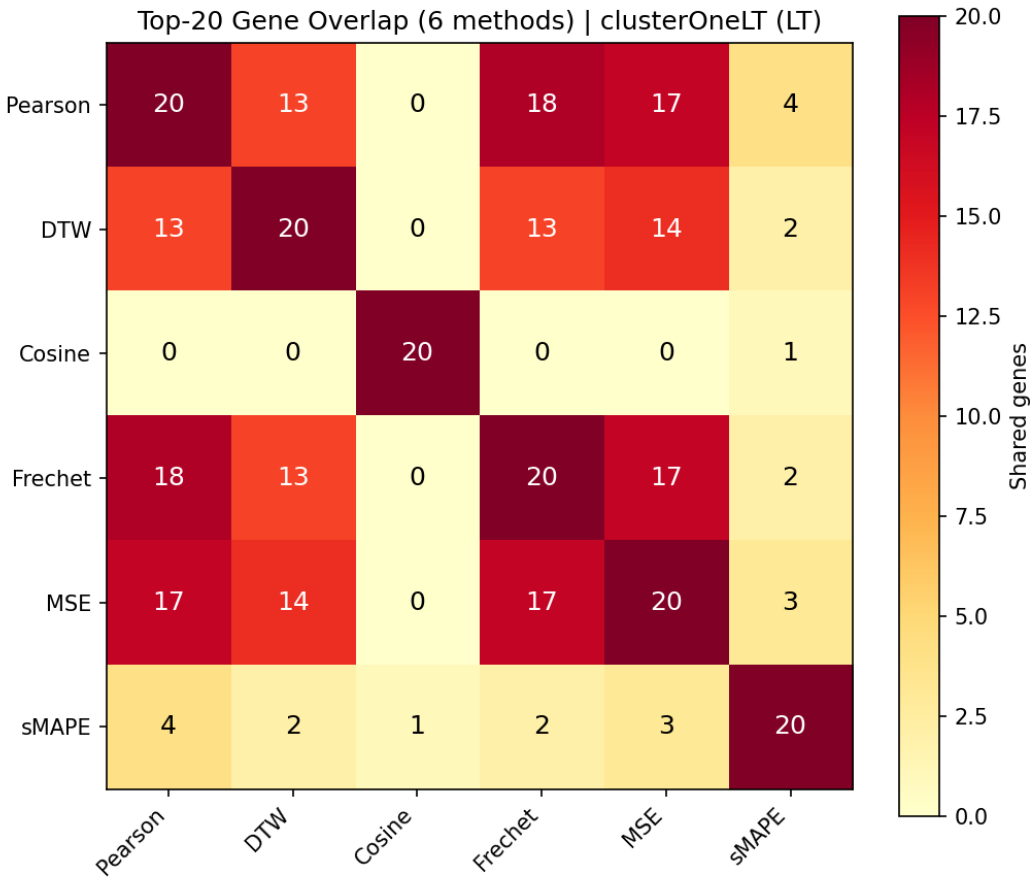
clusterOne LT — sMAPE



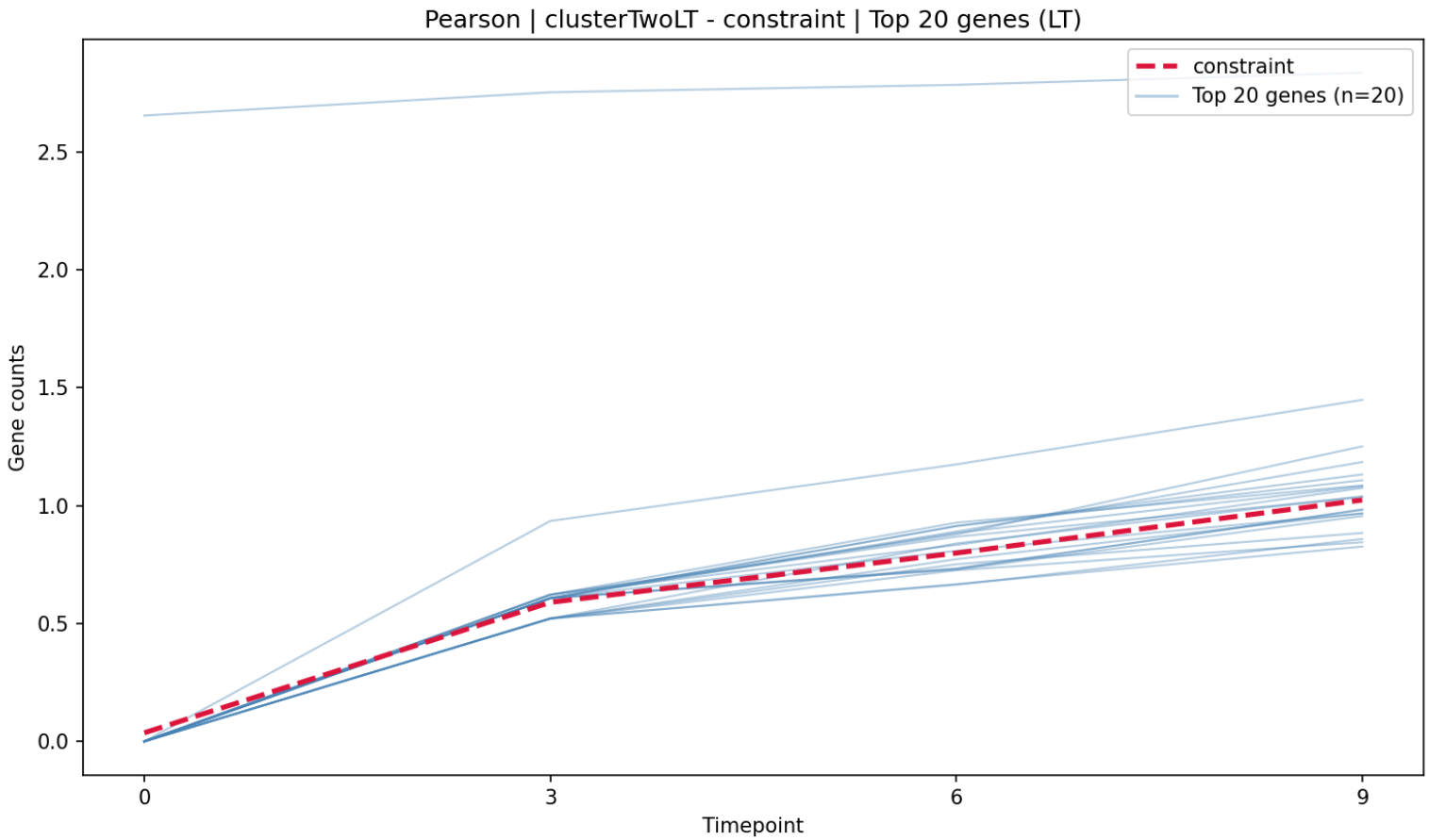
clusterOne LT — Ensemble



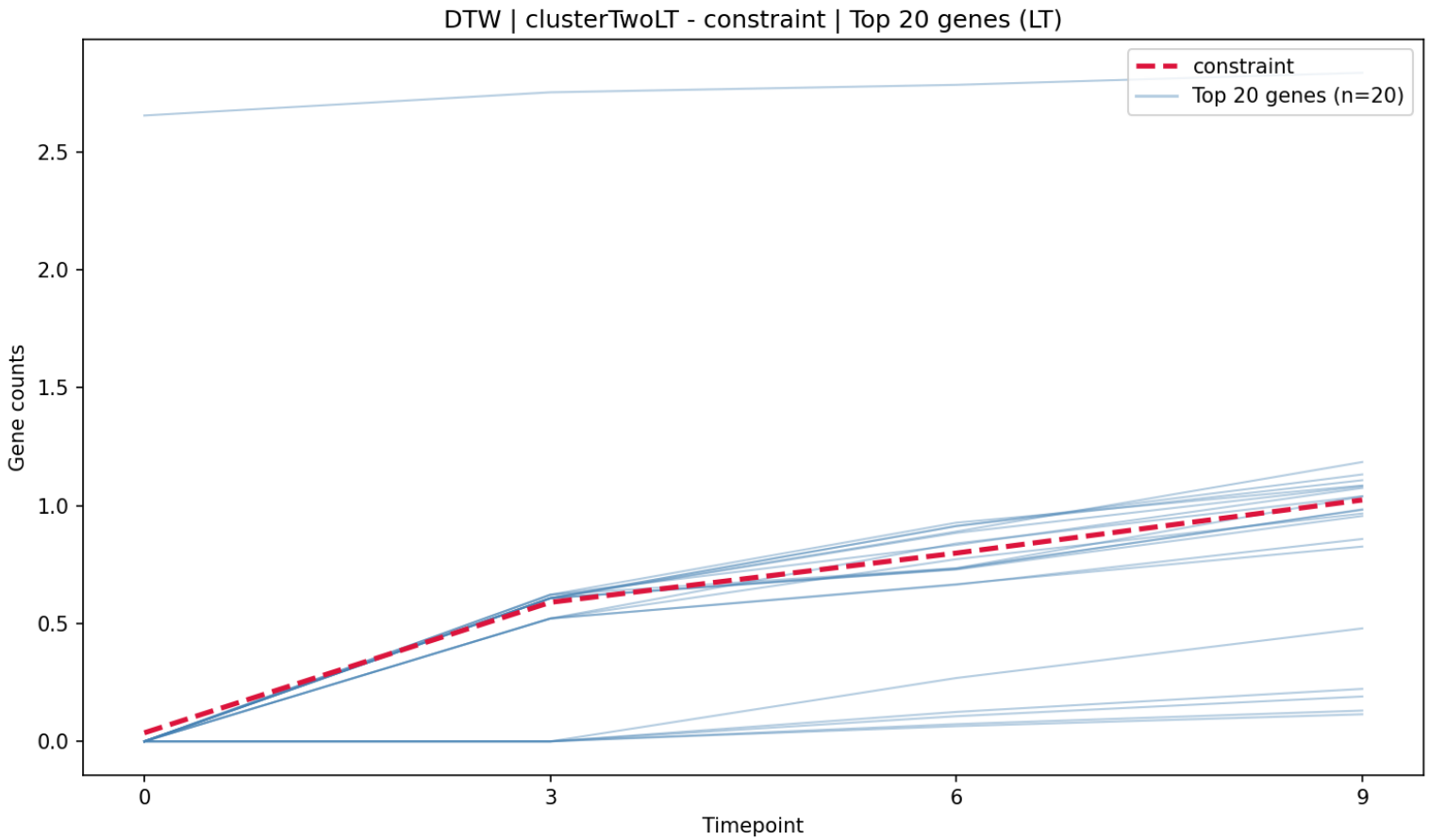
clusterOne LT — Overlap Heatmap



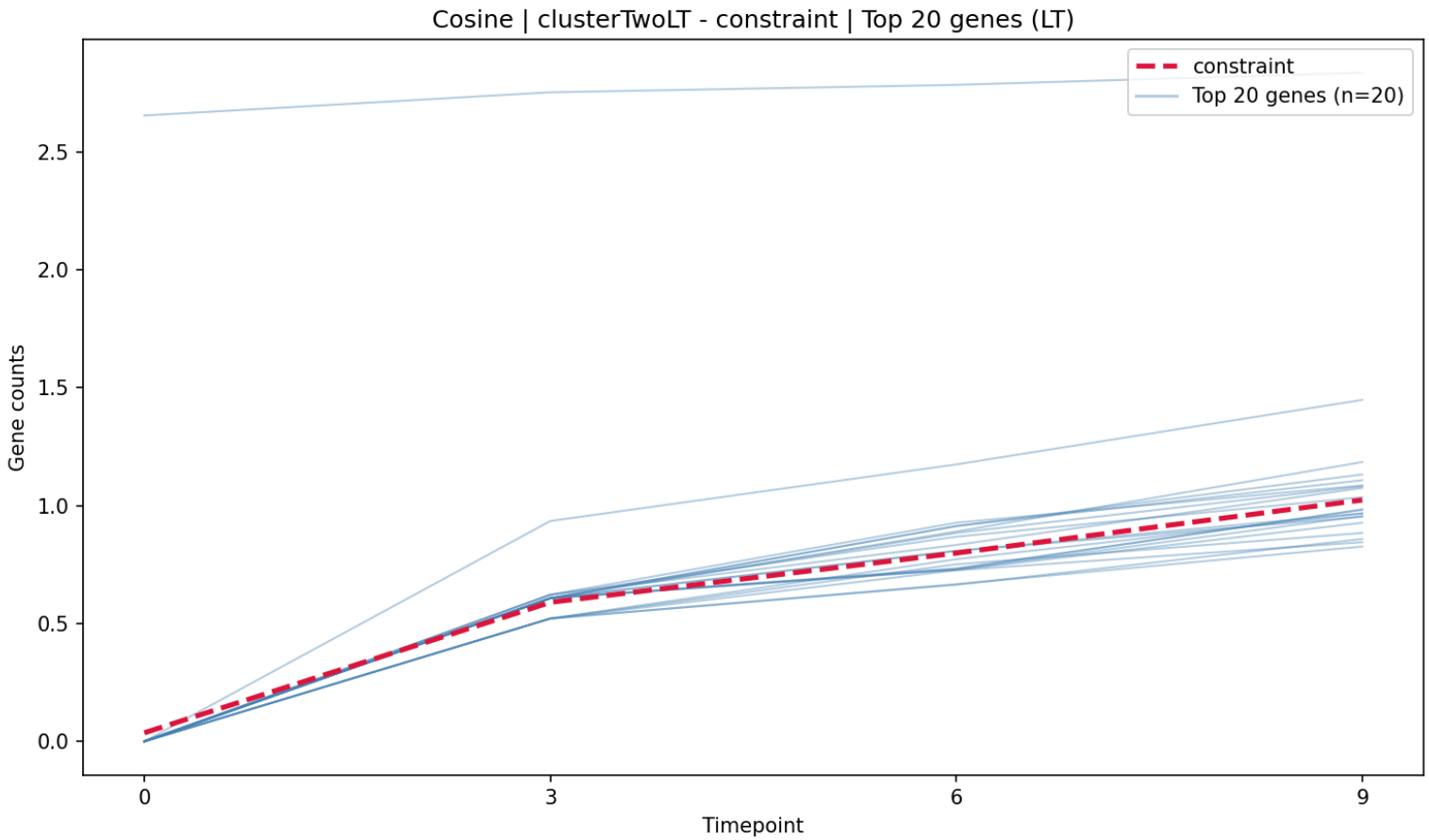
clusterTwo LT — Pearson



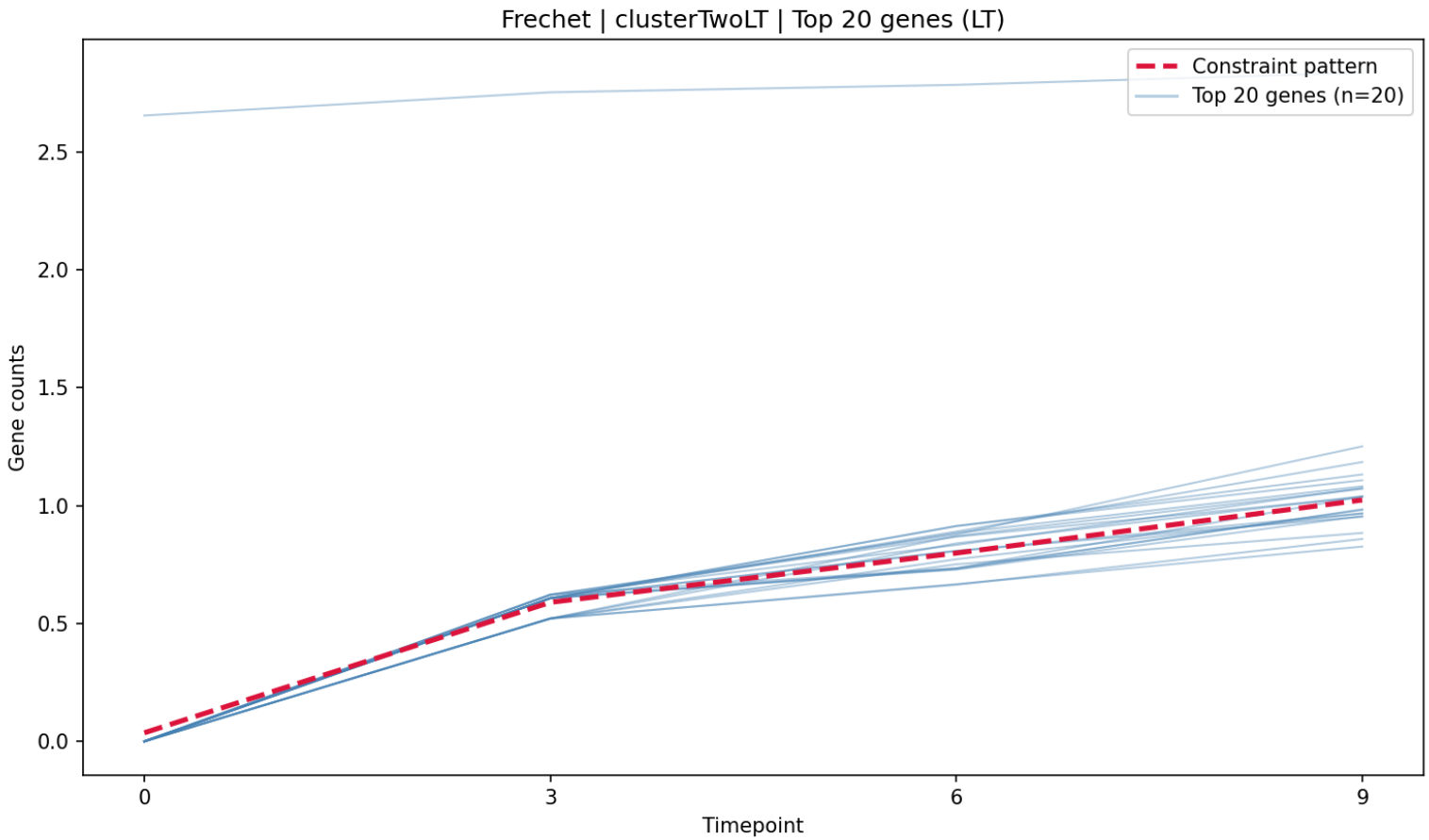
clusterTwo LT — DTW



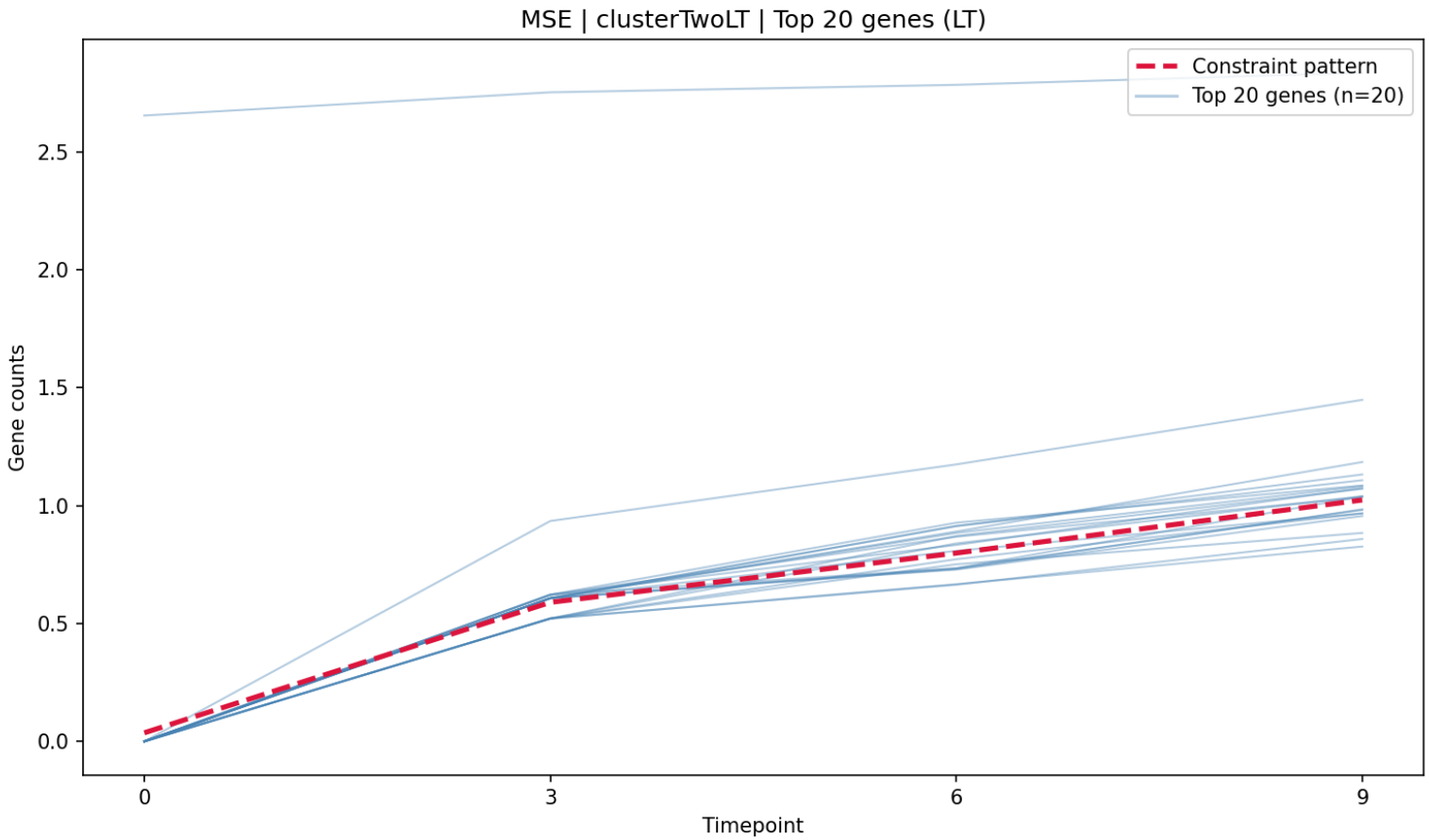
clusterTwo LT — Cosine



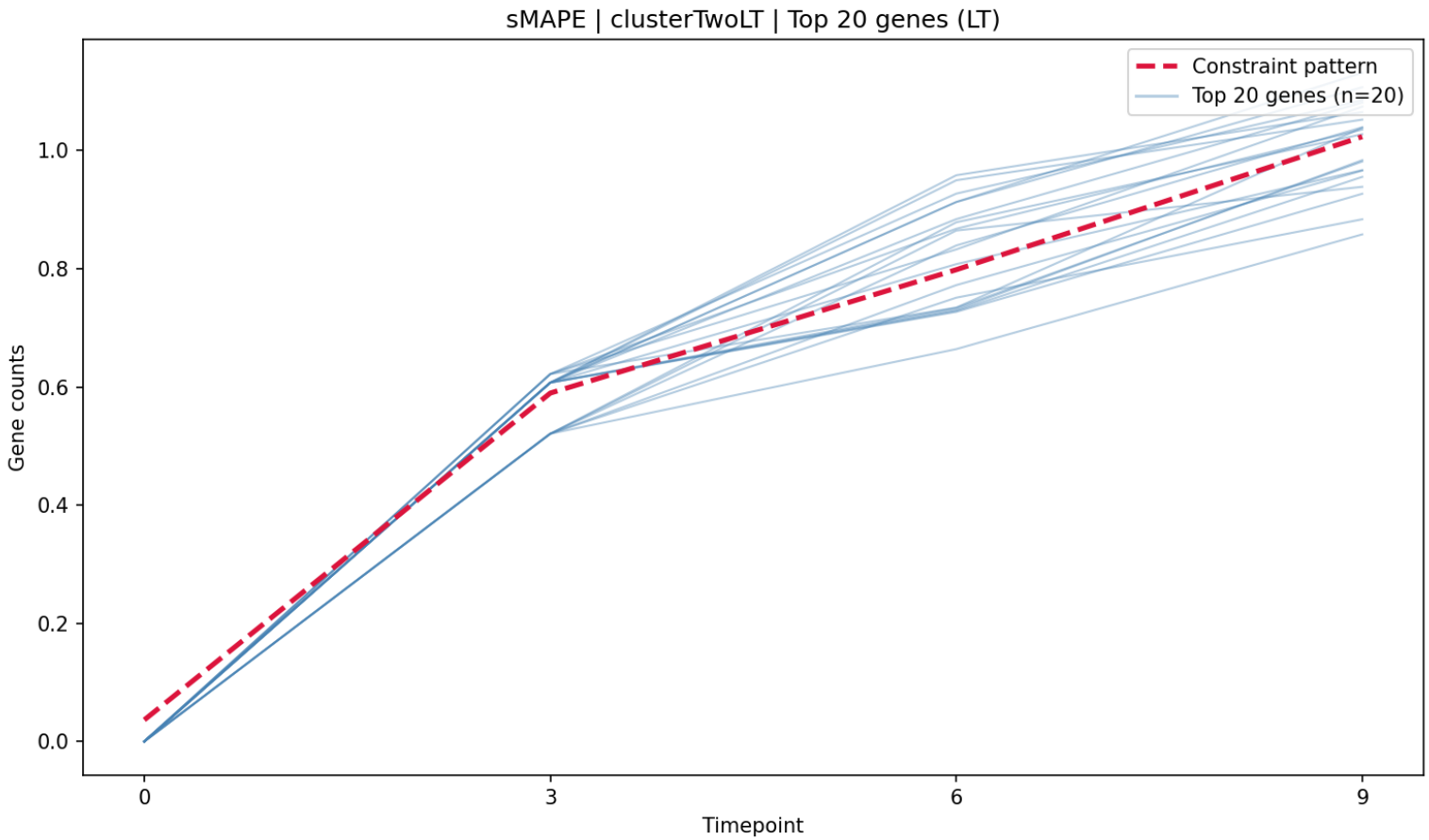
clusterTwo LT — Fréchet



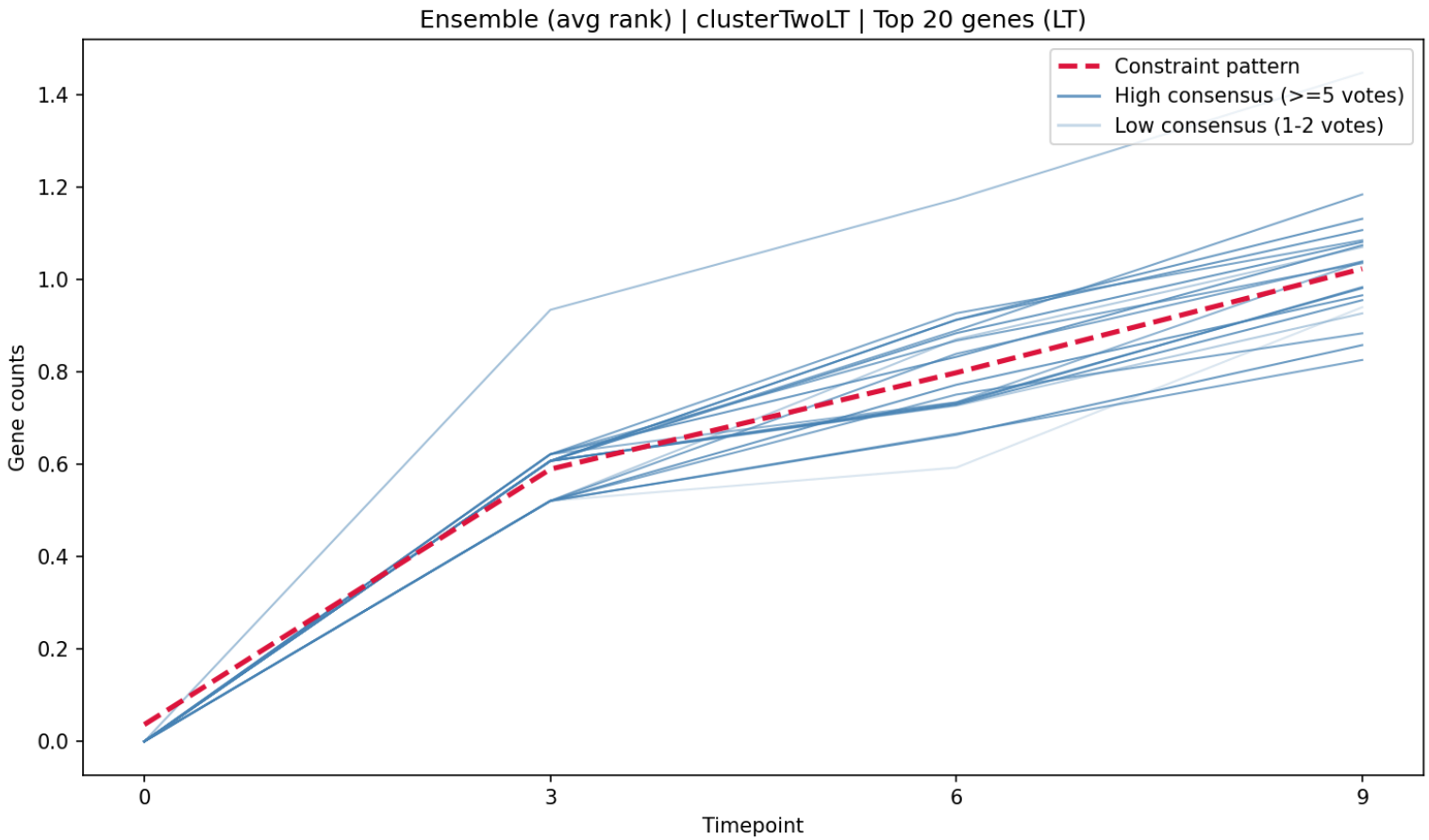
clusterTwo LT — MSE



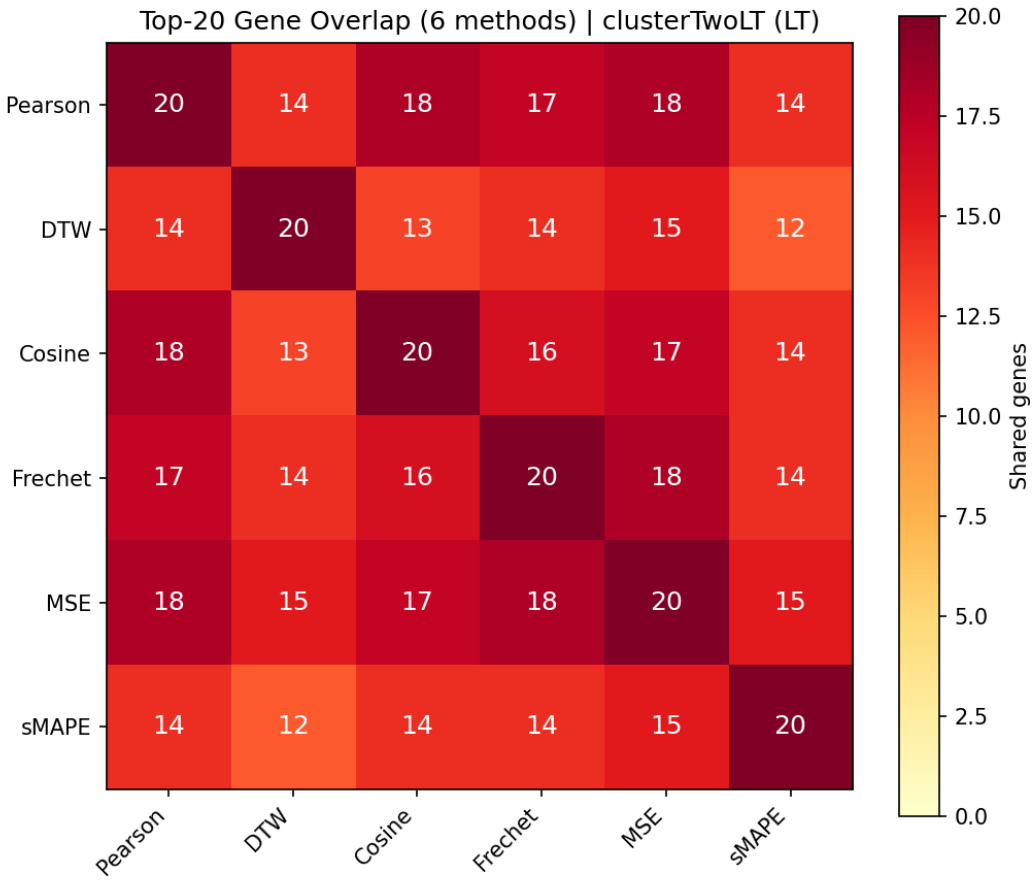
clusterTwo LT — sMAPE



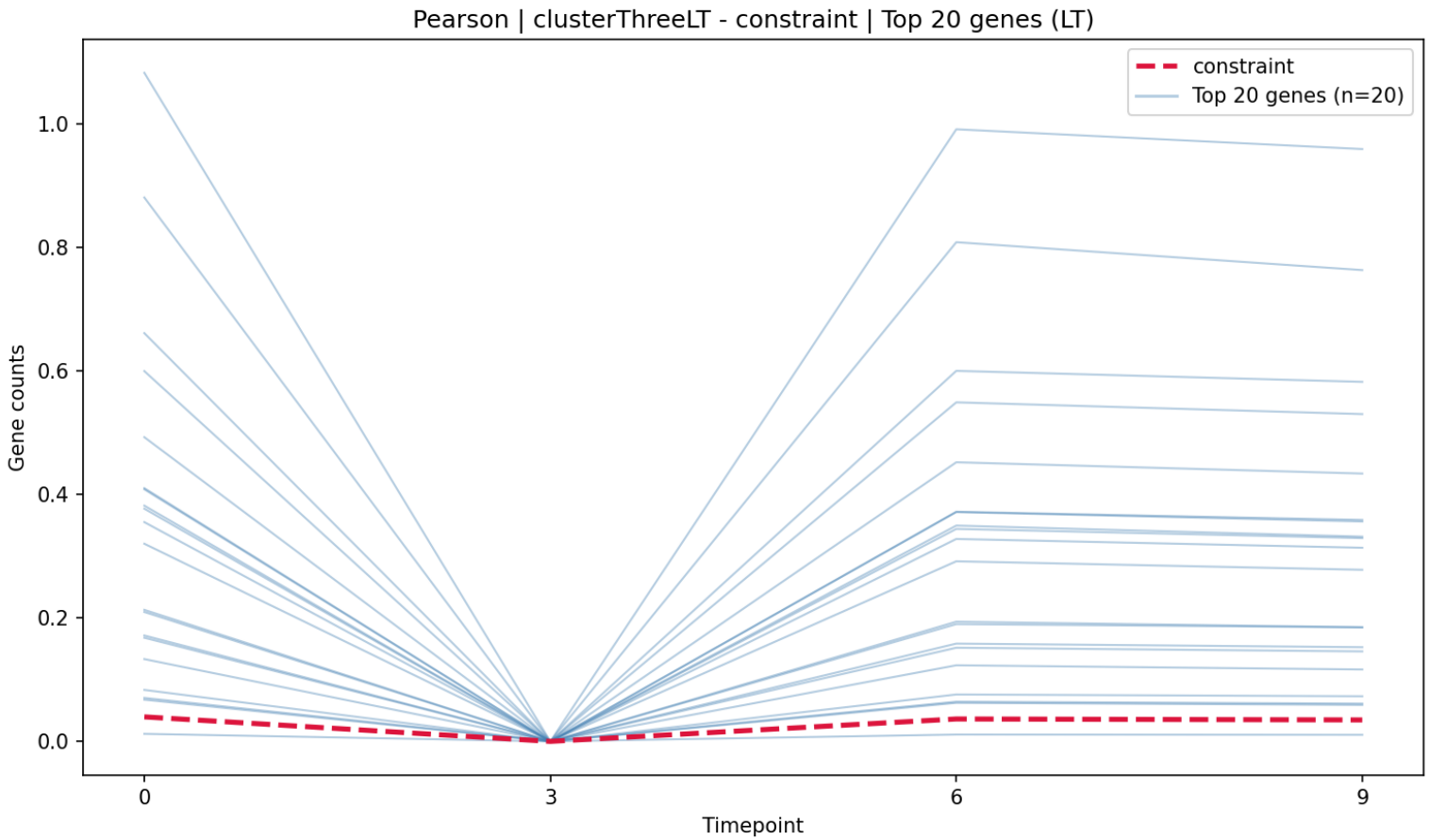
clusterTwo LT — Ensemble



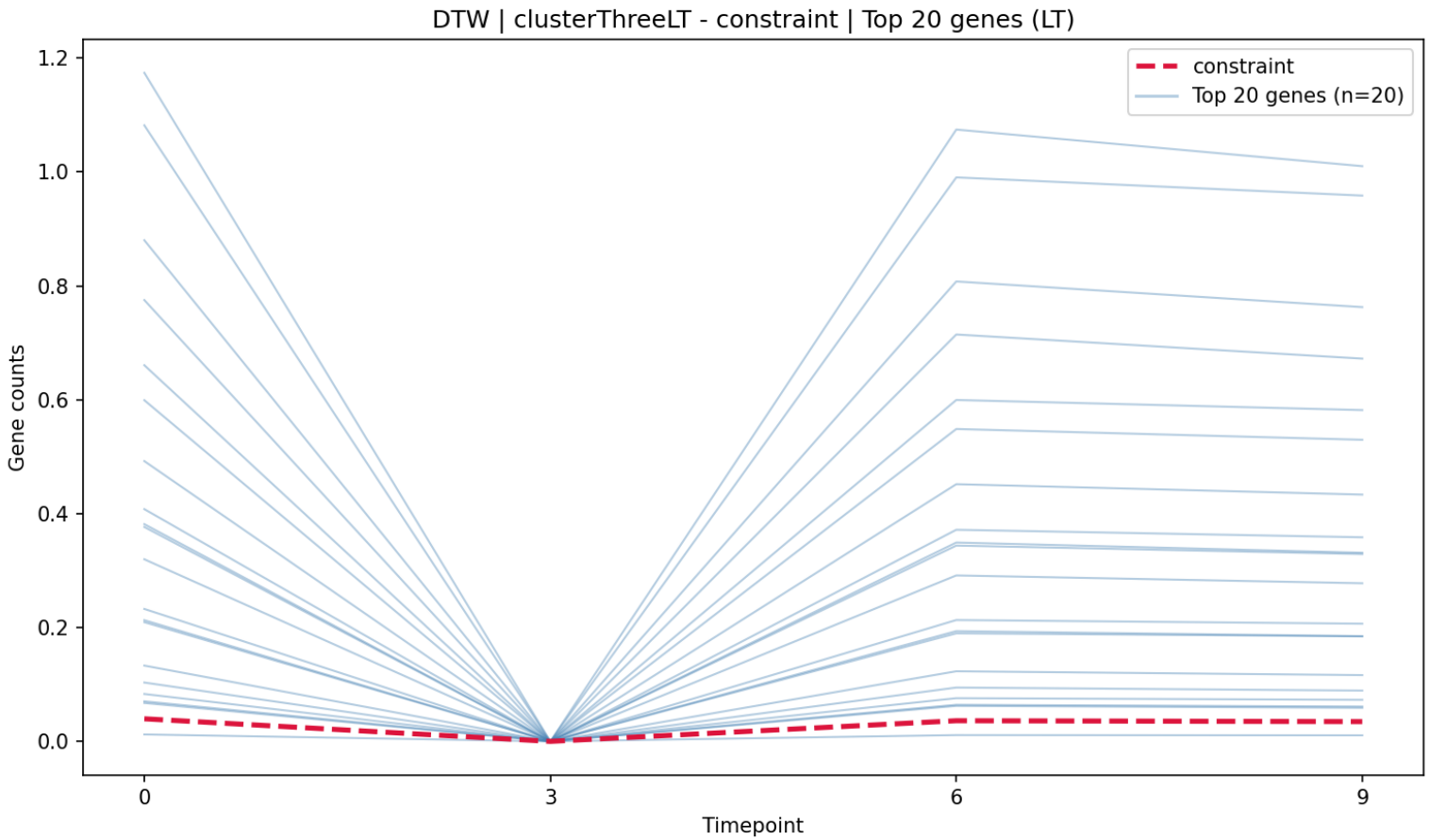
clusterTwo LT — Overlap Heatmap



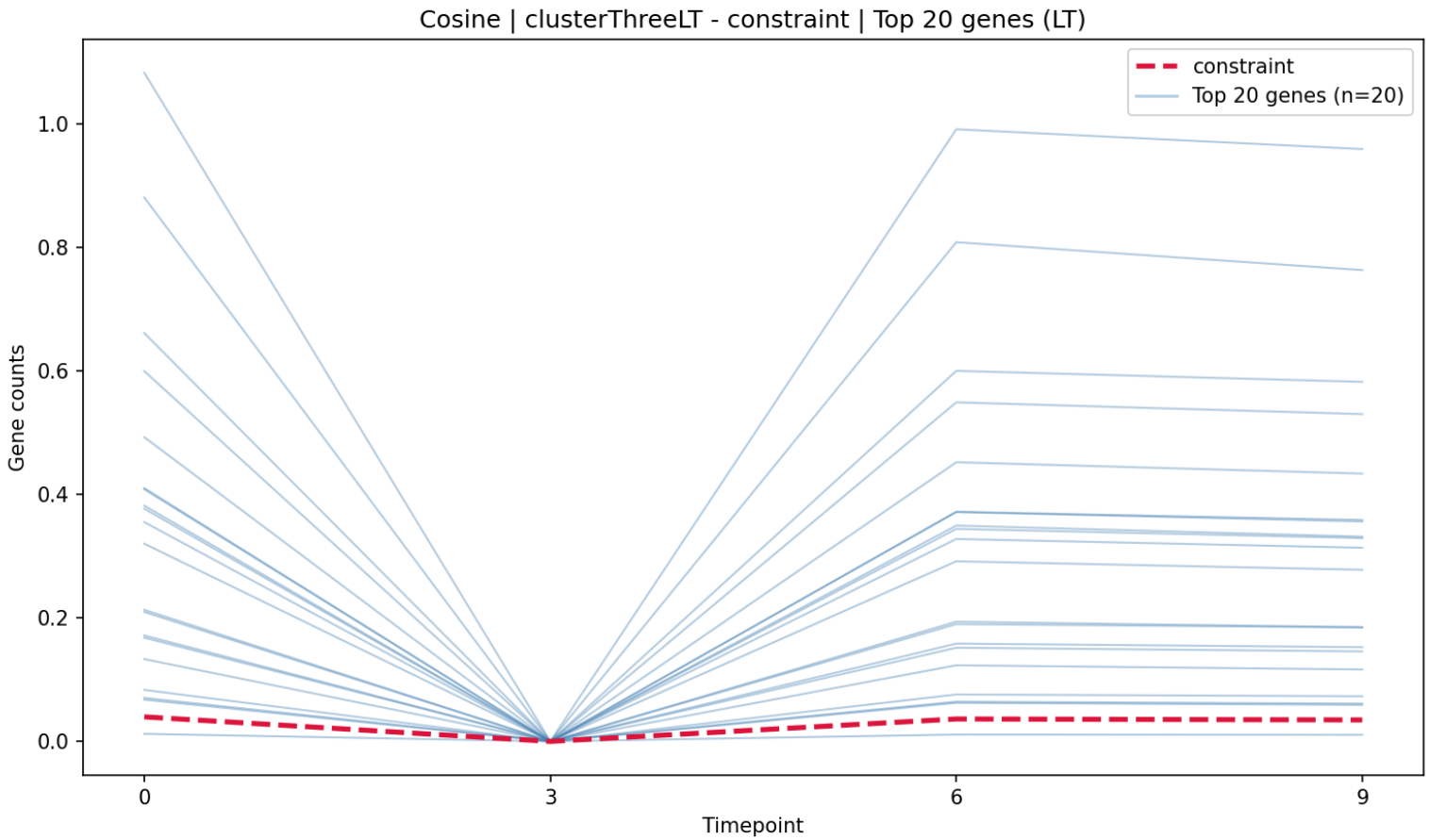
clusterThree LT — Pearson



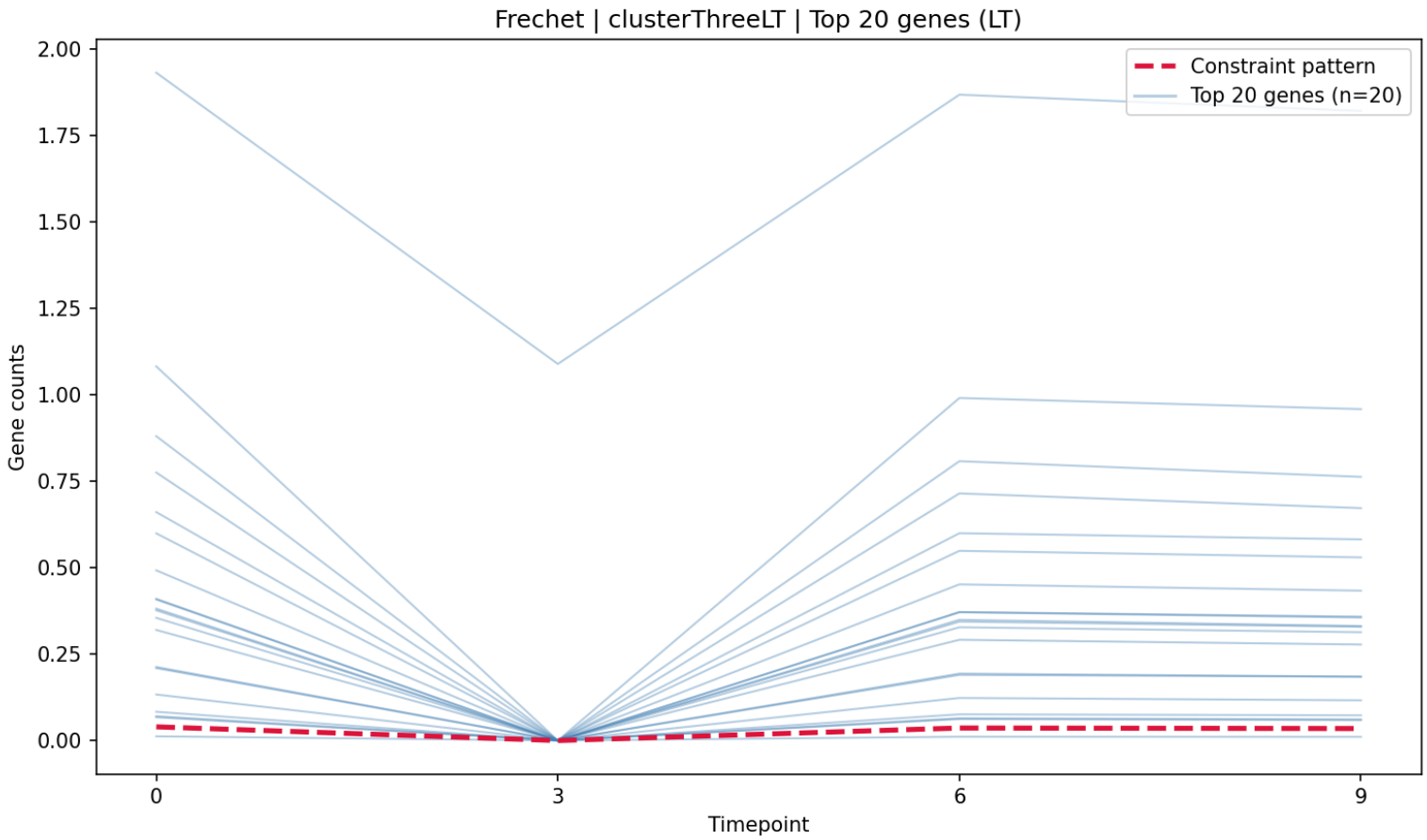
clusterThree LT — DTW



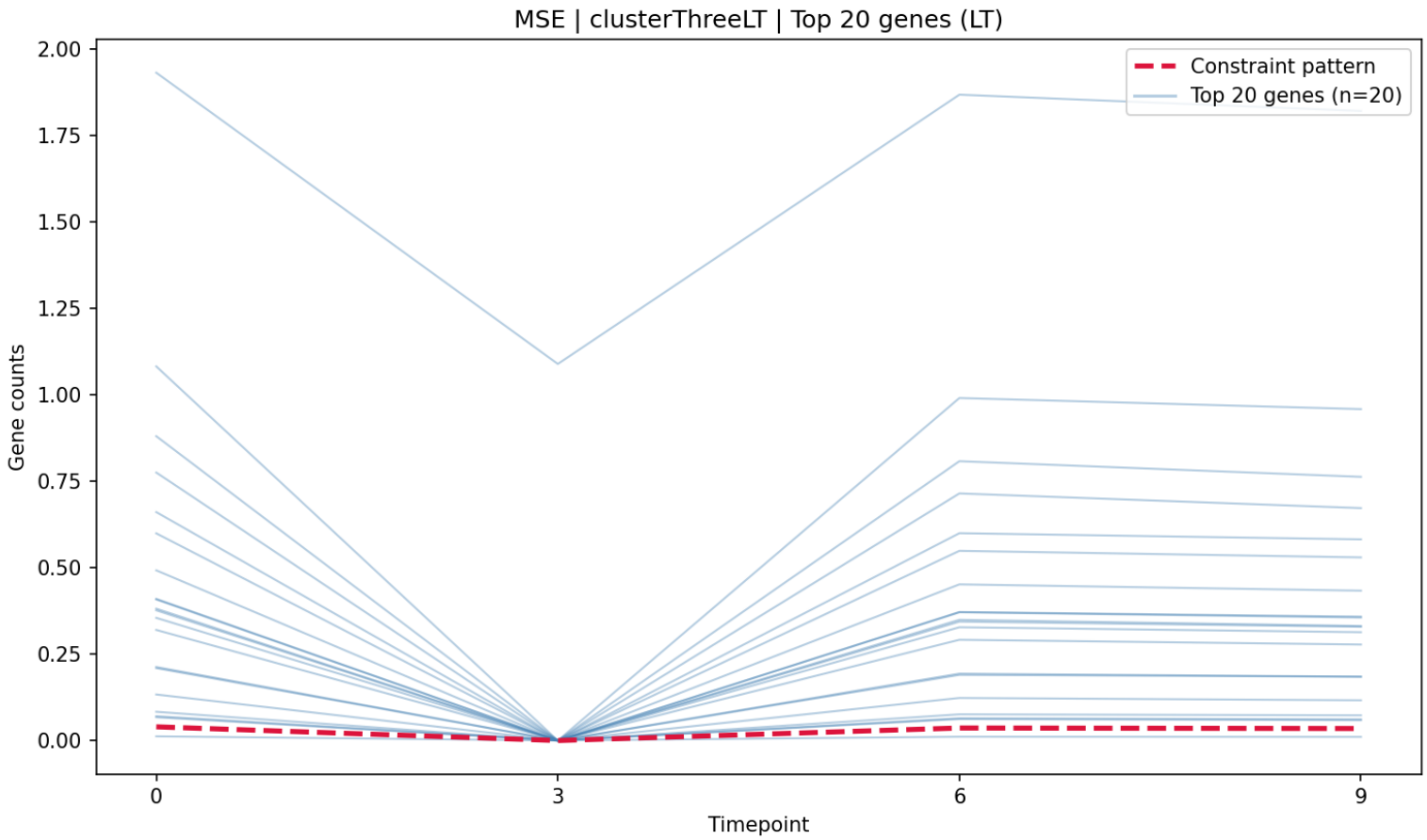
clusterThree LT — Cosine



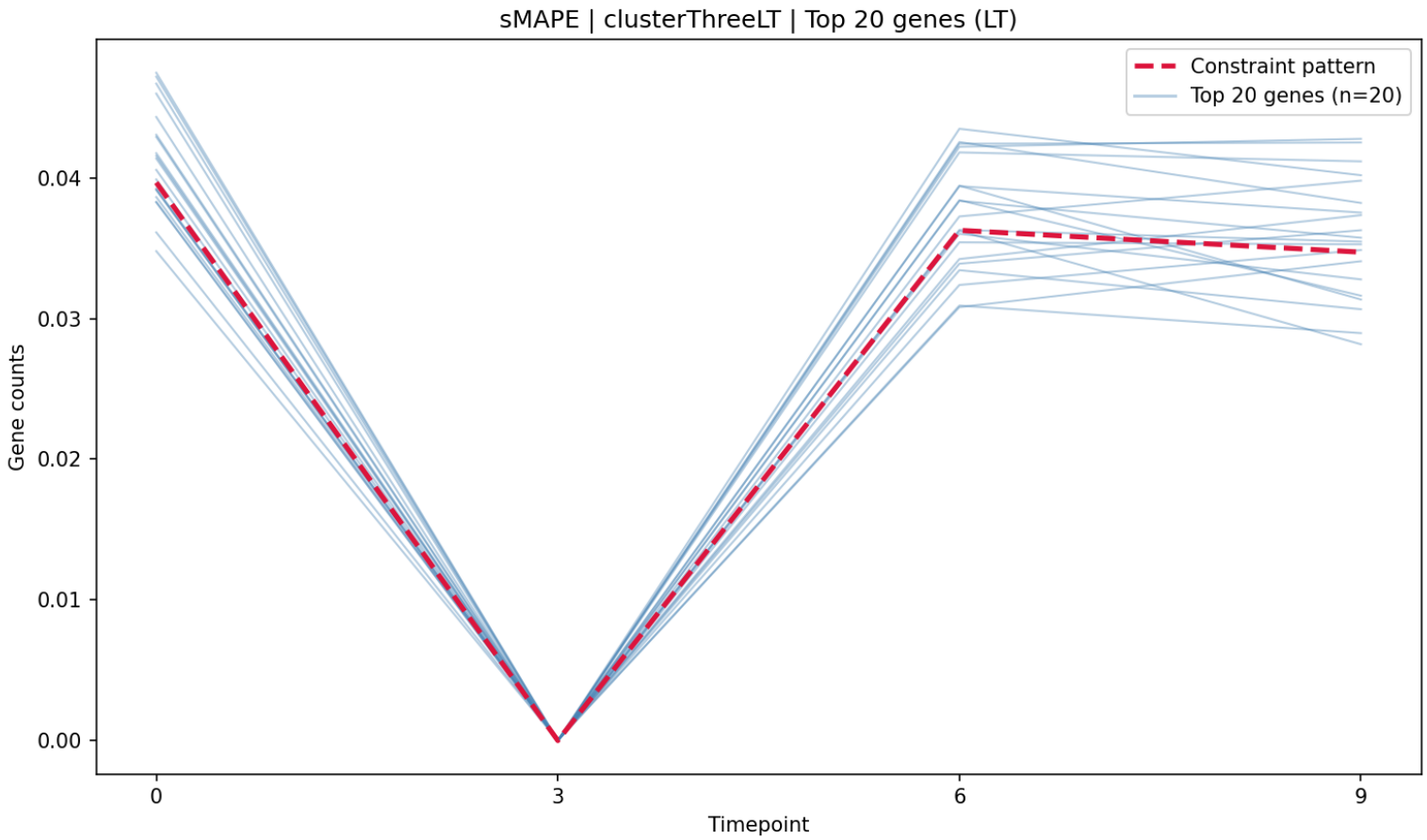
clusterThree LT — Fréchet



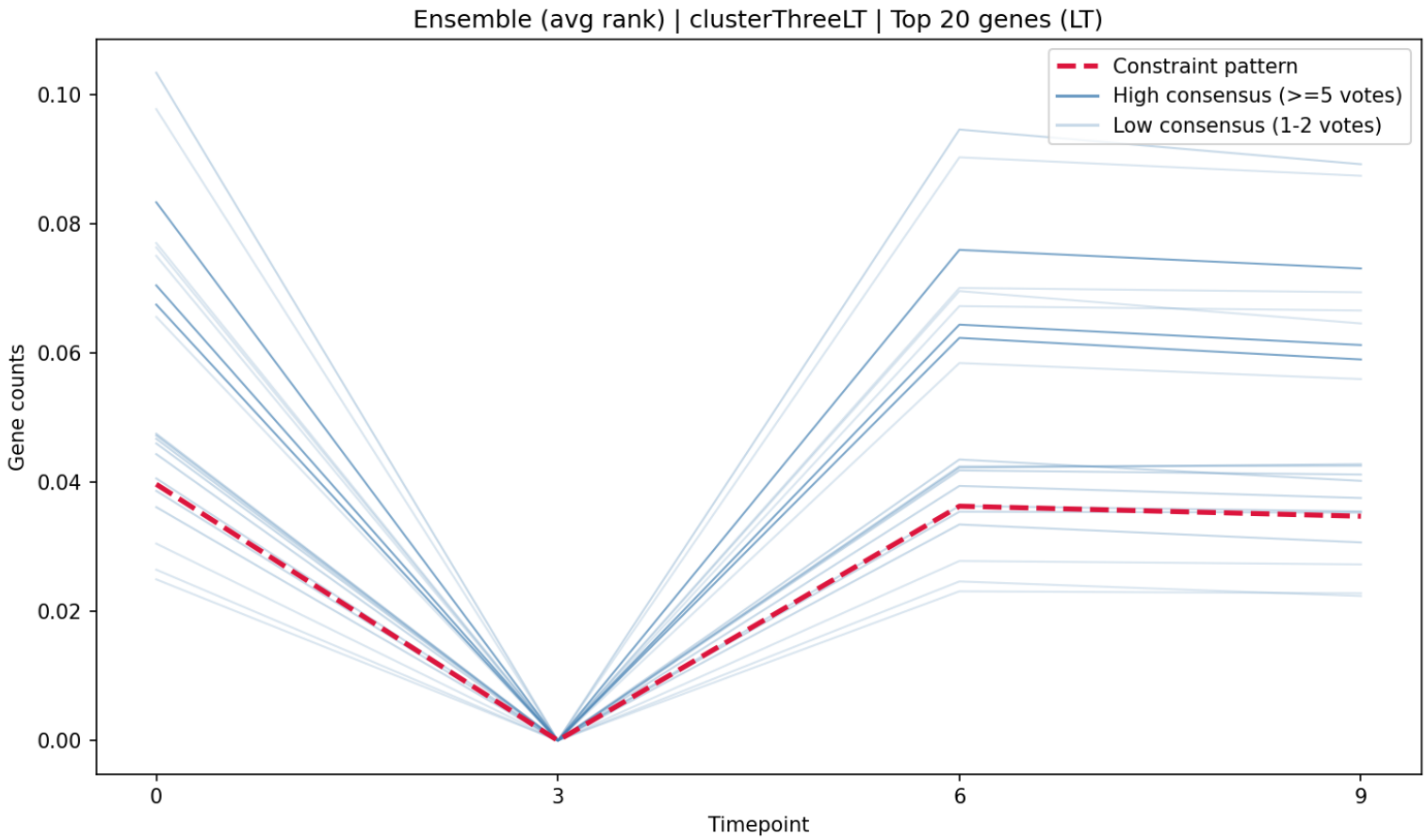
clusterThree LT — MSE



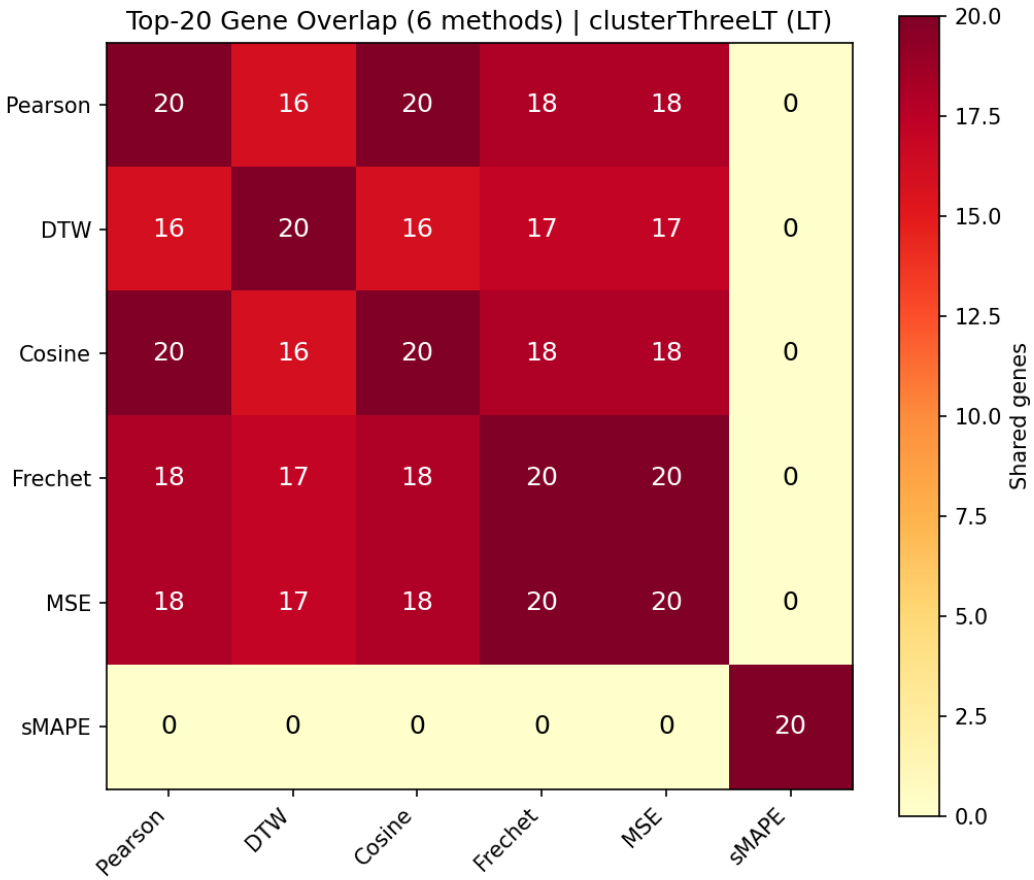
clusterThree LT — sMAPE



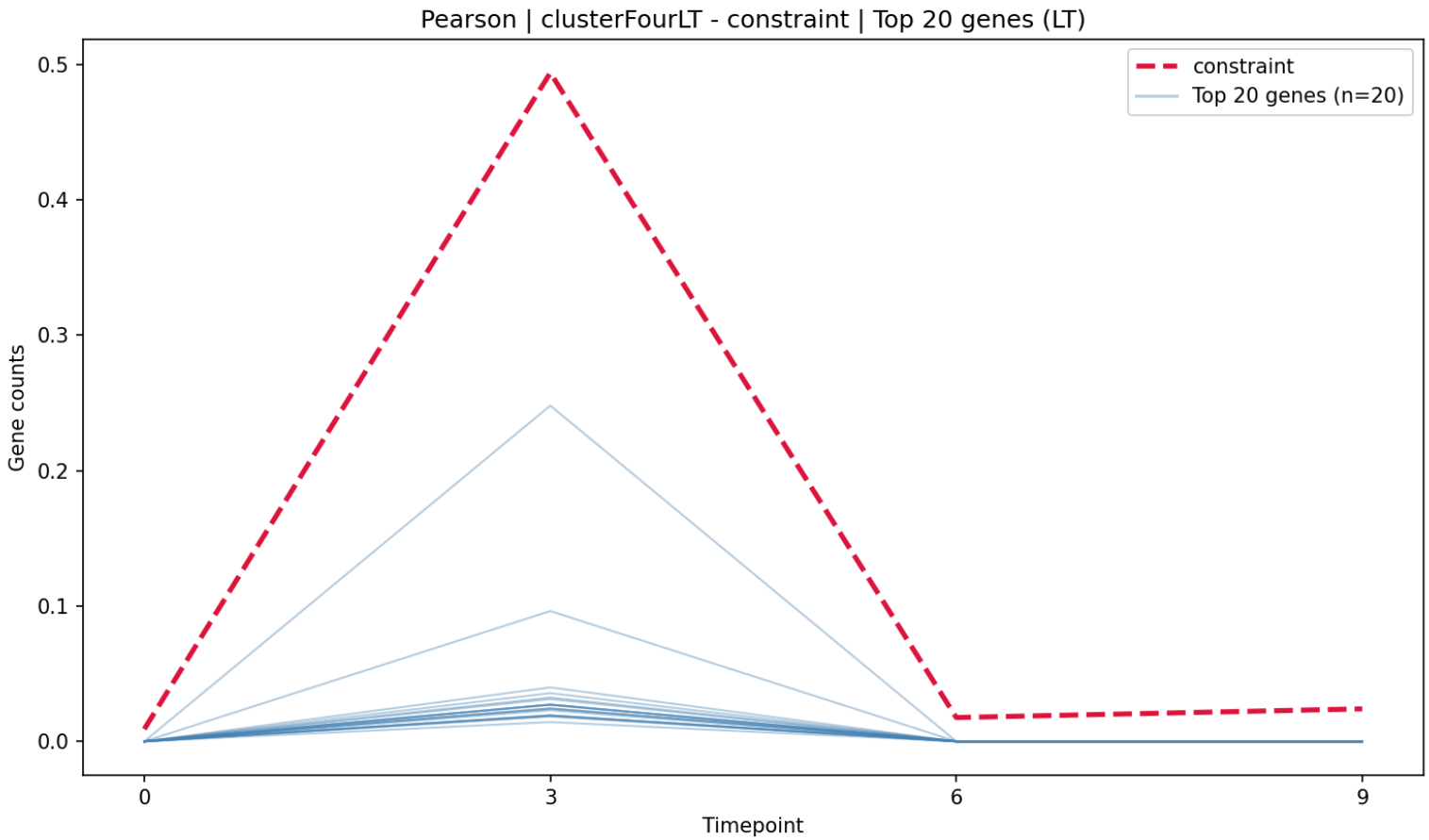
clusterThree LT — Ensemble



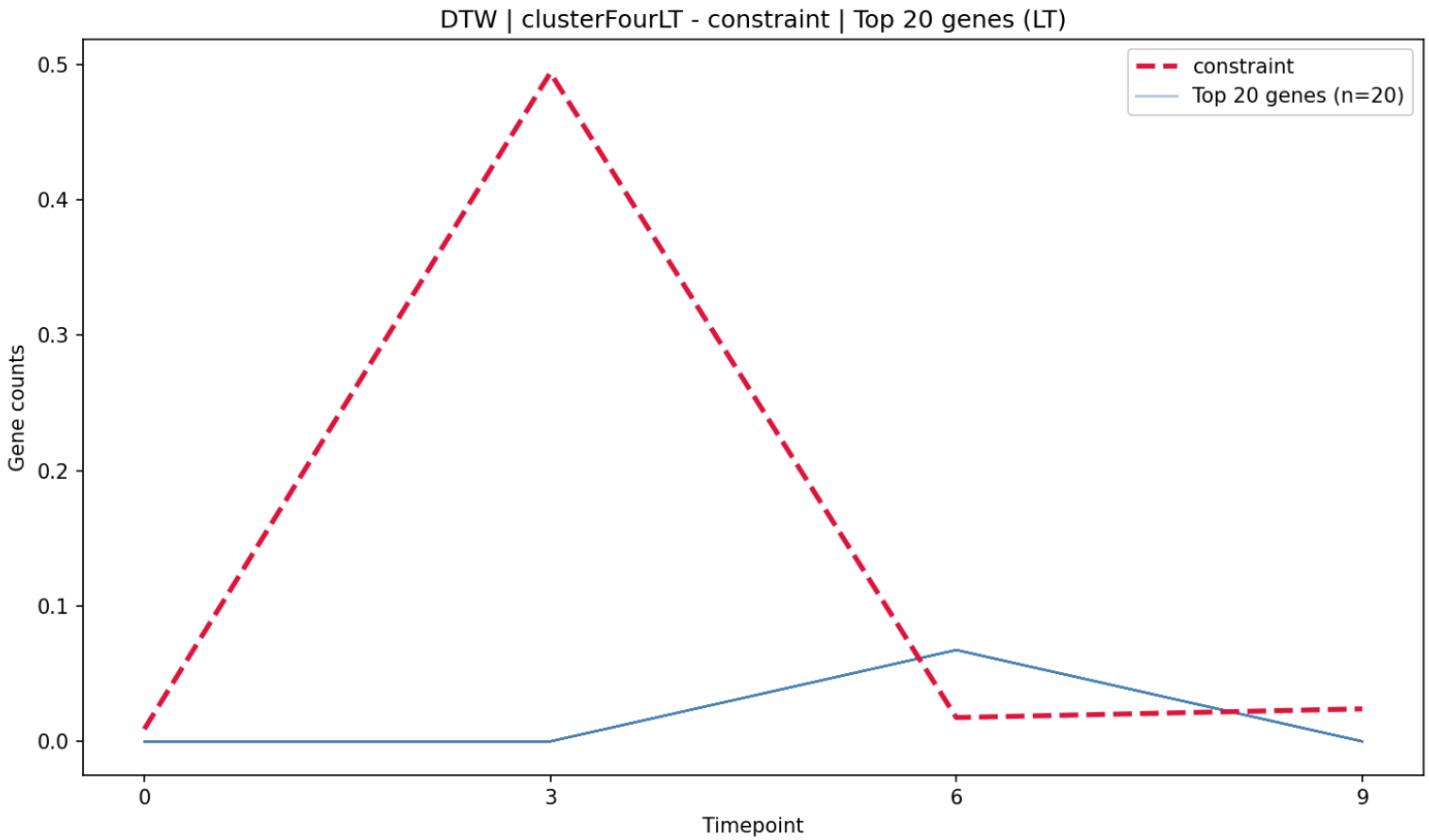
clusterThree LT — Overlap Heatmap



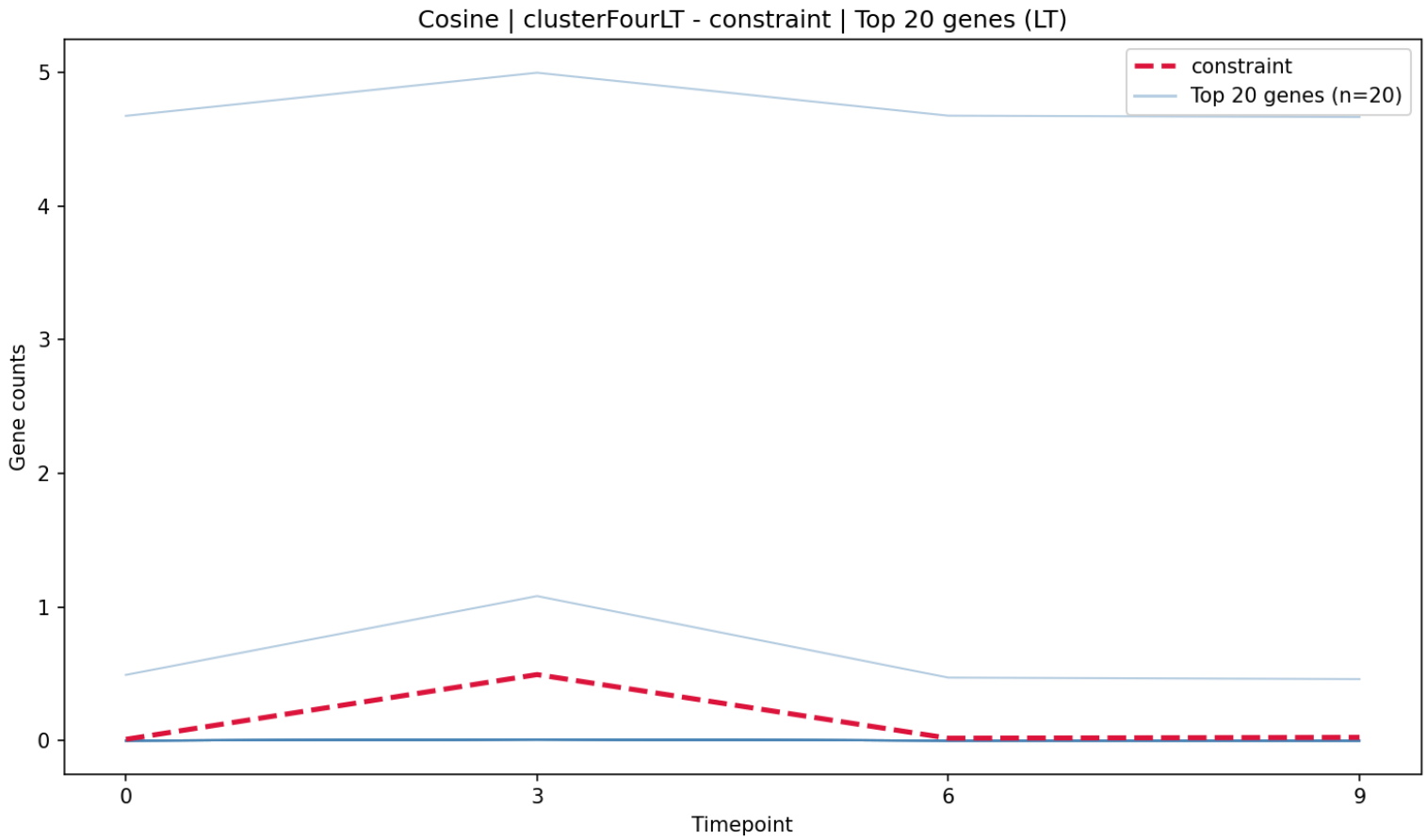
clusterFour LT — Pearson



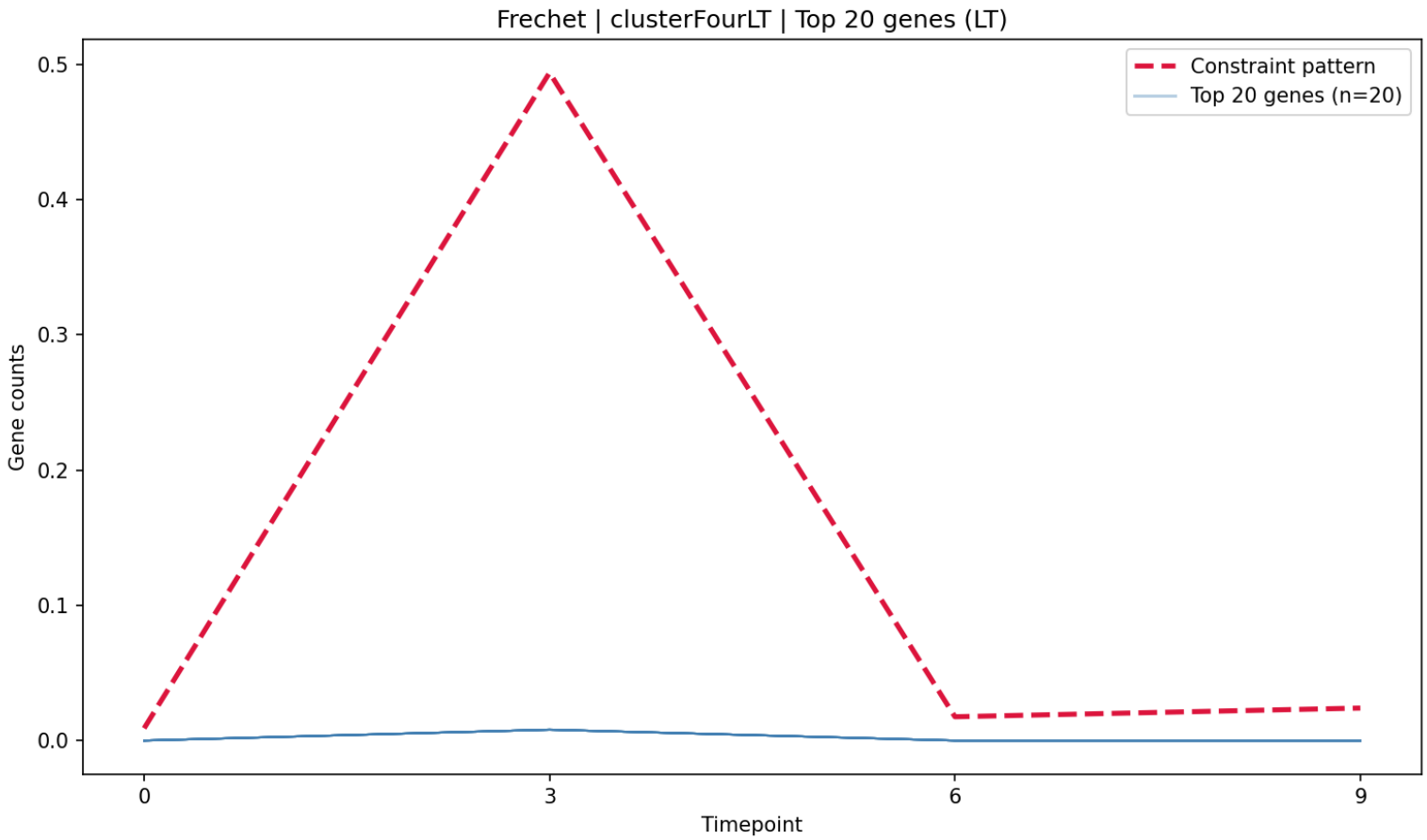
clusterFour LT — DTW



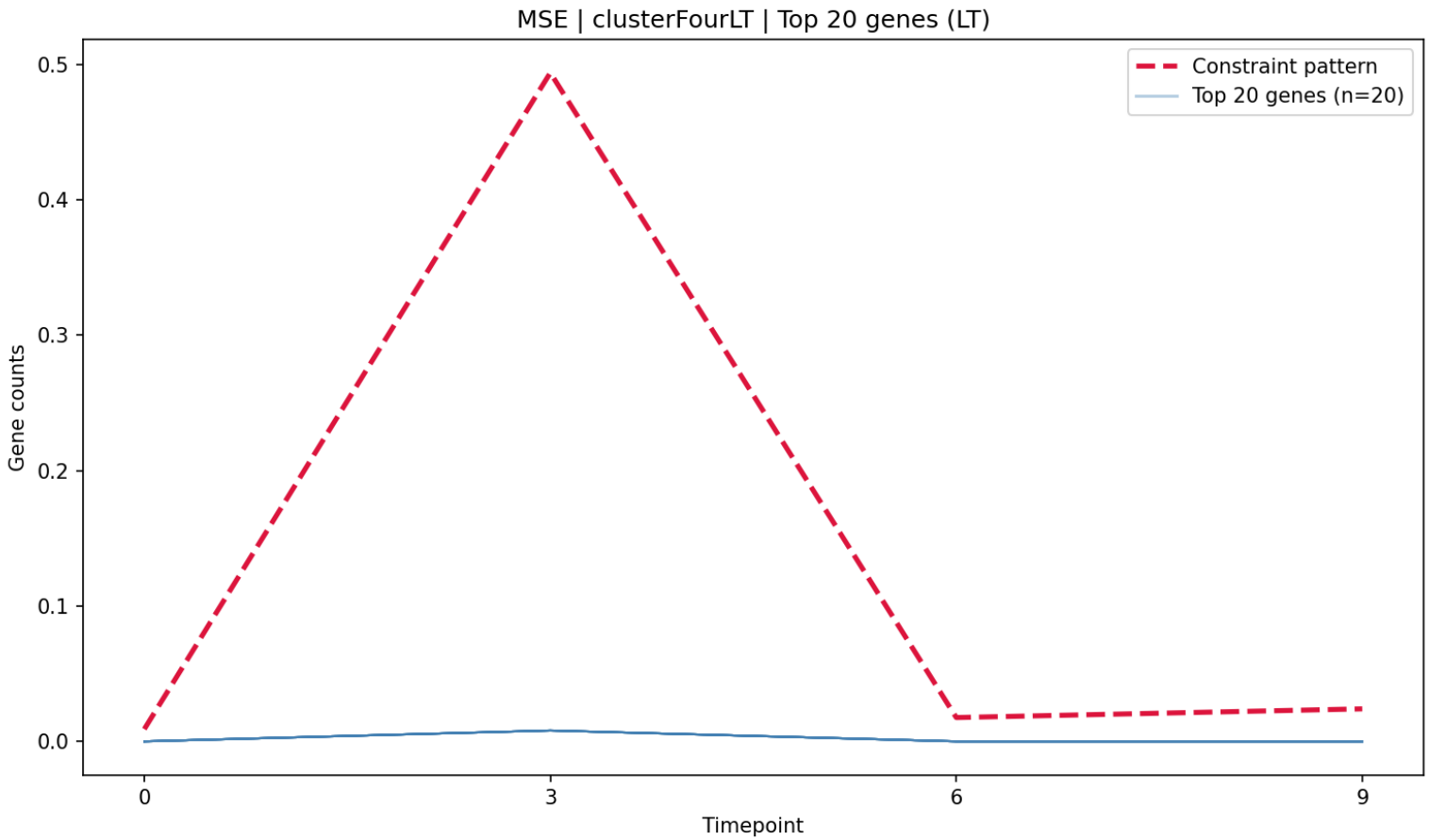
clusterFour LT — Cosine



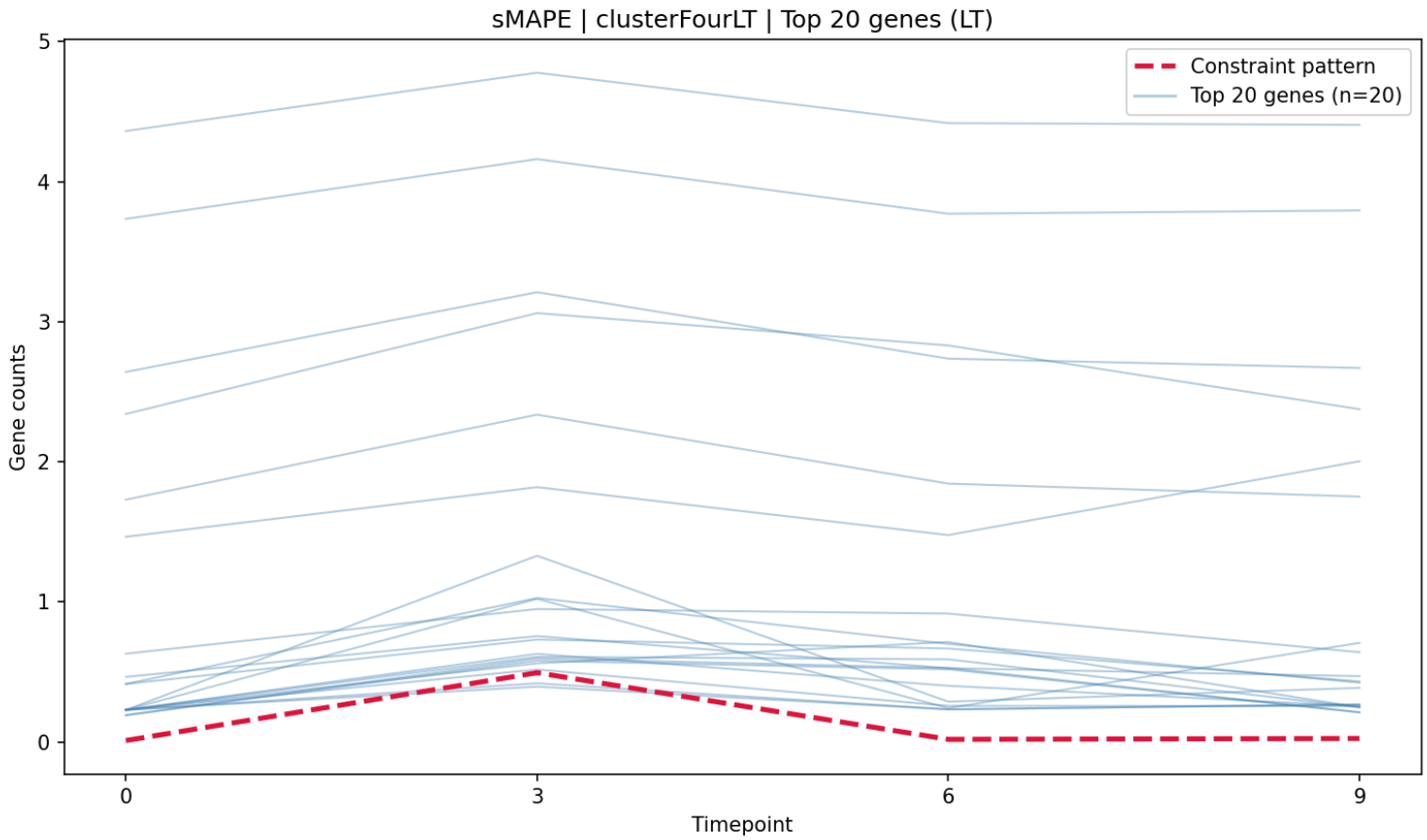
clusterFour LT — Fréchet



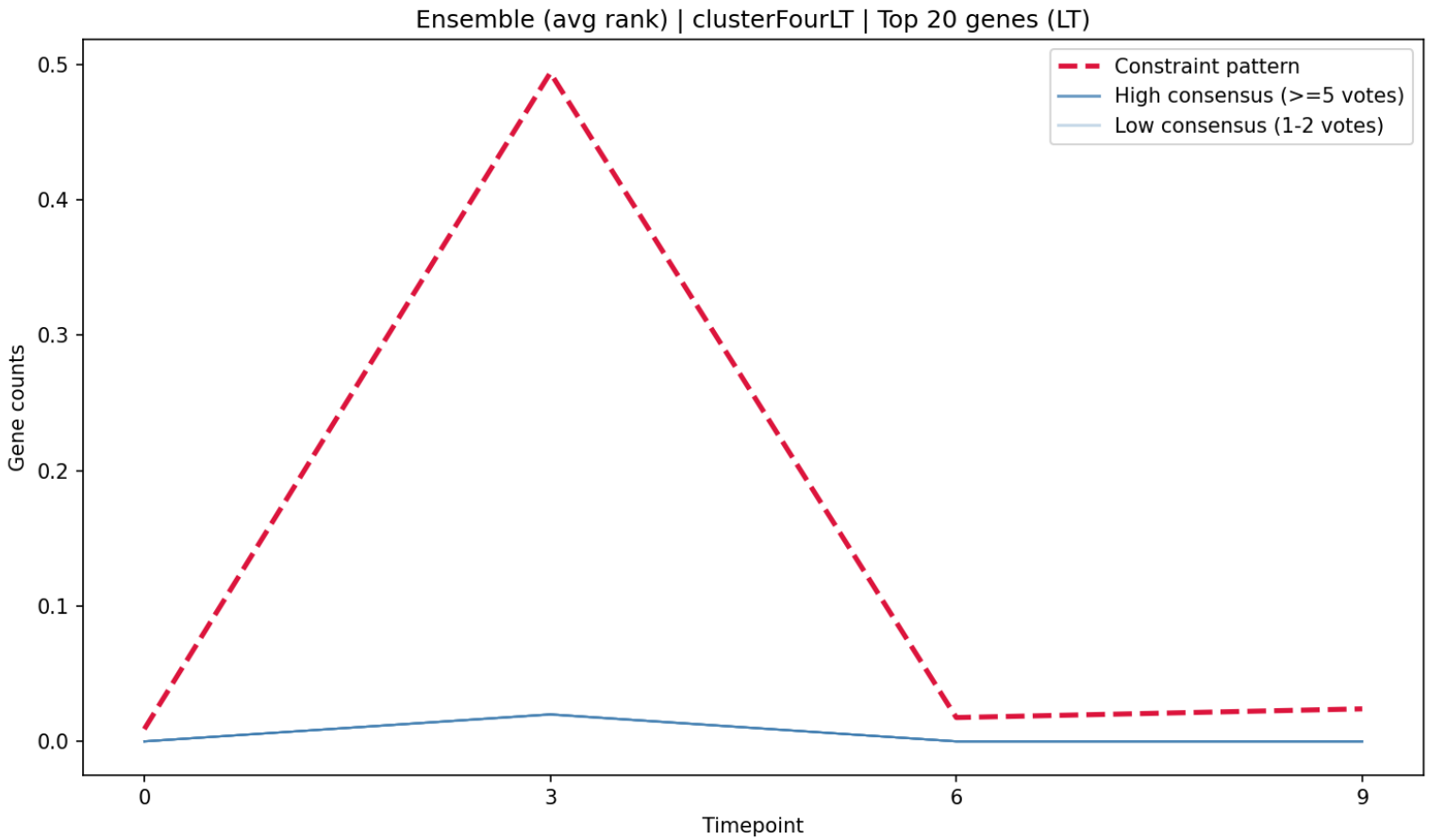
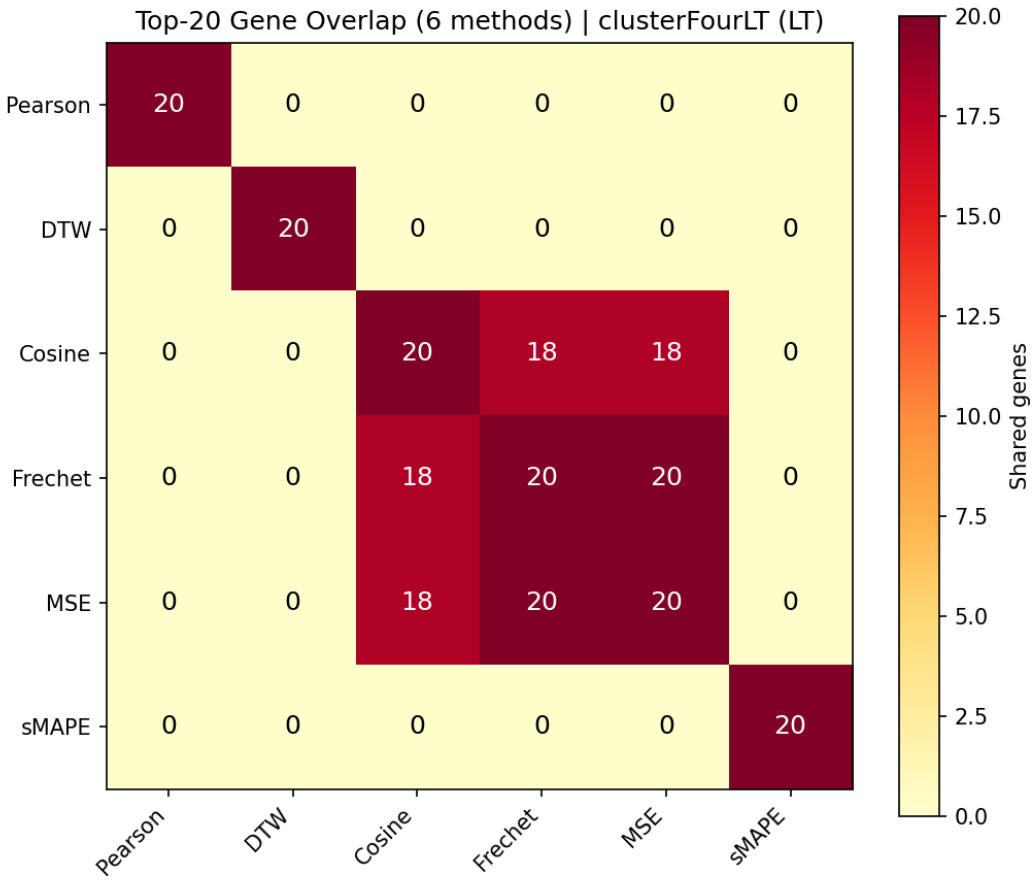
clusterFour LT — MSE



clusterFour LT — sMAPE



clusterFour LT — Ensemble



clusterFour LT — Overlap Heatmap